

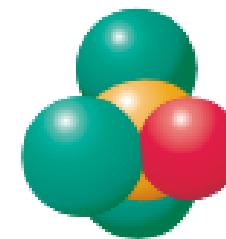
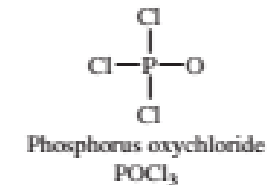
OLABISI ONABANJO UNIVERSITY
FACULTY OF SCIENCE
CHEMICAL SCIENCES DEPARTMENT

CHM 102: GENERAL CHEMISTRY II
(INORGANIC)

DR. SEGUN A. OGUNDARE

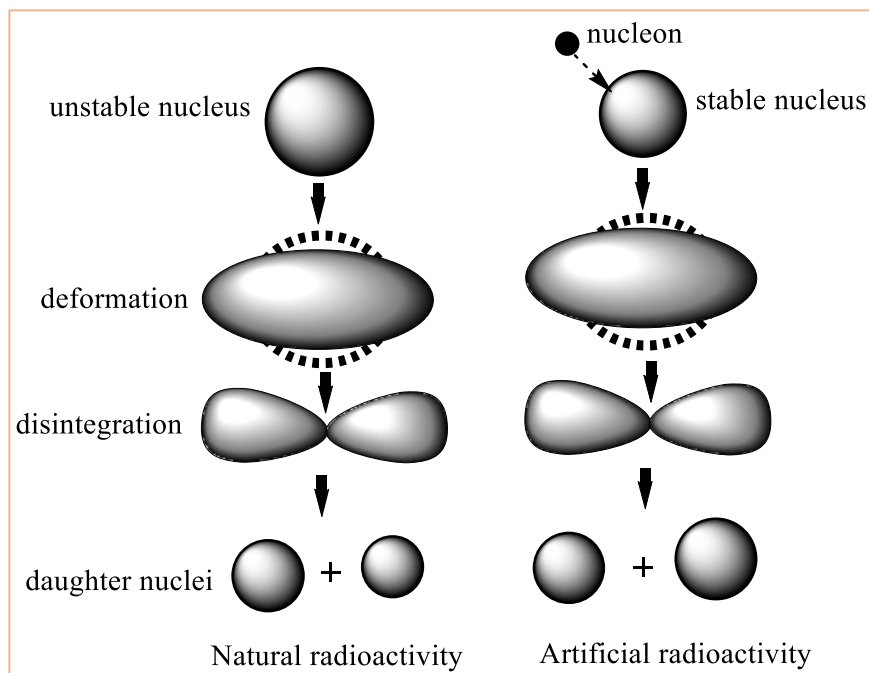
Content:

- **Bonding and Intermolecular Forces**
- **The Periodic Table and Periodicity**
- **Descriptive Inorganic chemistry of Main groups**
- **Brief introduction to transition metal Chemistry**
- **Nuclear Chemistry**



1 H																	2 He
3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
55 Cs	56 Ba	57 La	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
87 Fr	88 Ra	89 Ac	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Uun								

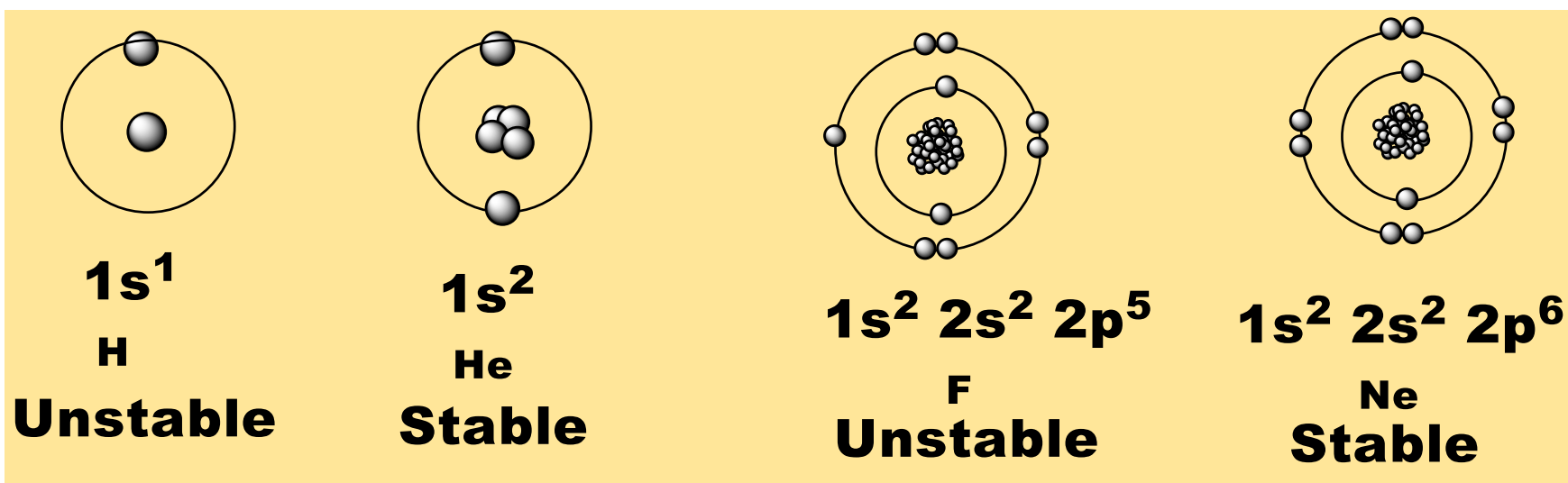
58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu
90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr



Bonding and Intermolecular Forces

Chemical Bonding

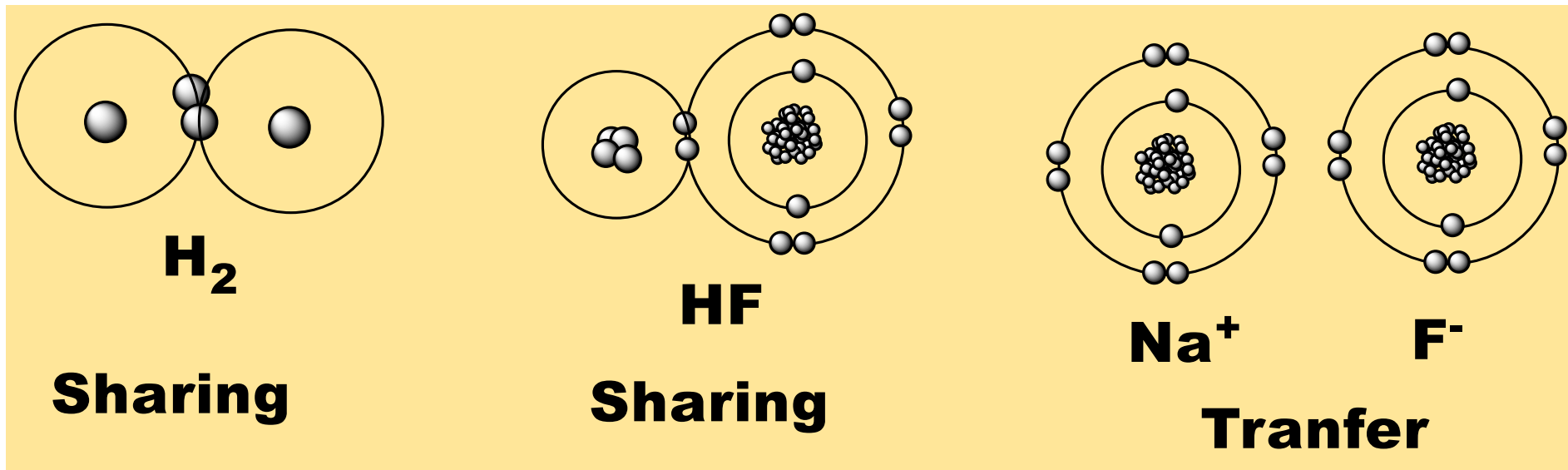
- Chemical bonding is an interaction between atoms of the same or different elements with the purpose of attaining stable electronic configuration. The electronic configuration is the arrangement or distribution of electrons in the atomic orbitals.
- A stable electronic configuration is attained if the shells are filled with electrons.
- The outermost or the valence shell should have 8 electrons with the exception of the first period elements, which are H and He.



Bonding and Intermolecular Forces (*contd.*)

Chemical Bonding

- According to G. N. Lewis an atom is most stable if its outermost shell is either filled or contains eight electrons.
- An atom will lose, gain, or share electron(s) in order to achieve a stable electronic configuration of the nearest noble gas.
- The tendency of an atom to attain an electronic configuration, such that its valence shell contains 8 electrons is known as the **octet rule**.

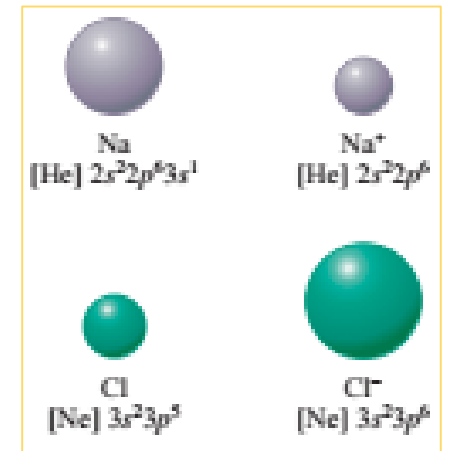
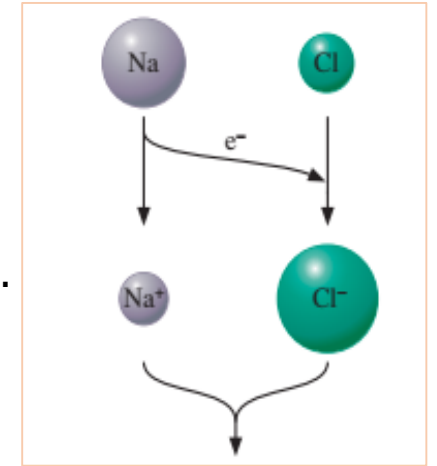


Bonding and Intermolecular Forces (*contd.*)

Types of Chemical Bonds: The ionic (electrovalent) and the covalent bonds.

The Ionic (electrovalent) Bond

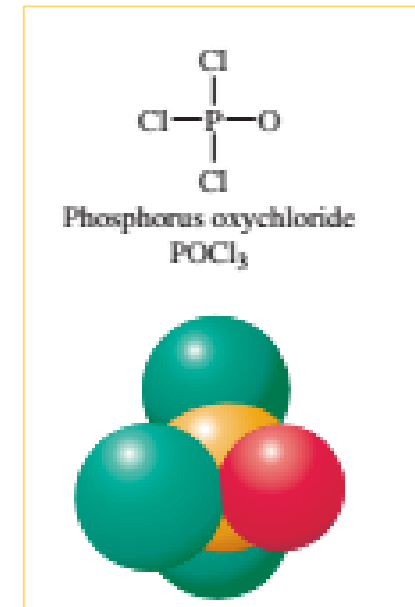
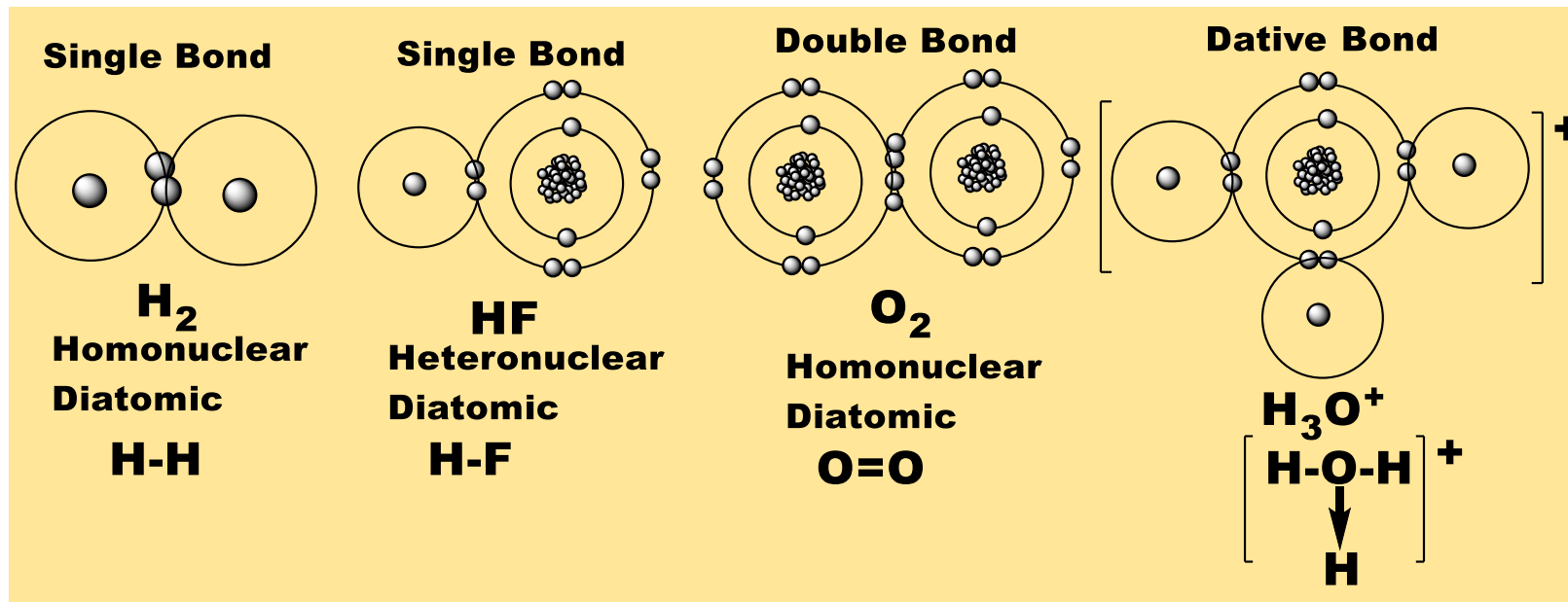
- This is a description of a chemical bond that involves loss or gain of electron(s) by atoms.
- The atom that loses electron(s) will have a decrease in its number of electrons.
- the atom that accepts the lost electron(s) will have an increase in number of its electrons.
- The atom that lost electron(s) transforms to positive (cation) ion
- The atom that gained electron(s) transforms to negative (anion) ion.



Bonding and Intermolecular Forces (*contd.*)

The Covalent Bond (Ordinary and Dative)

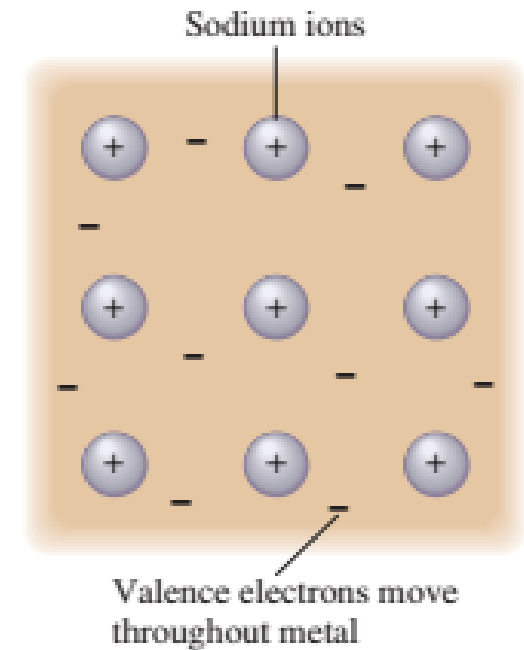
- This is a description of a chemical bond that involves sharing of pair(s) of electrons.
- An Ordinary covalent bond is formed when the shared pair is equally donated by the atoms.
- A Dative or coordinate covalent bond is formed when the shared pair is donated by one of the atoms.



Bonding and Intermolecular Forces (*contd.*)

Other types of bonds:

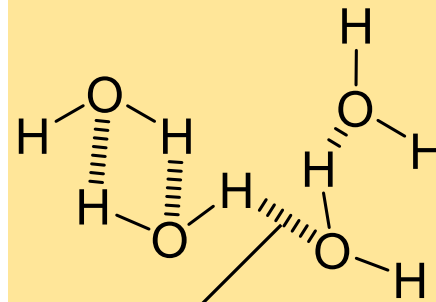
- **Metallic bond:** It exists between metallic atoms of the same element or alloys.
- Metal valence electrons are not strongly attracted to the nucleus.
- The electrons move from one atom to another linking up to form conducting band on the metal surface.
- The bond exists in solid metals like aluminum, iron and copper.
- The higher the valence electrons the greater the metallic bond.



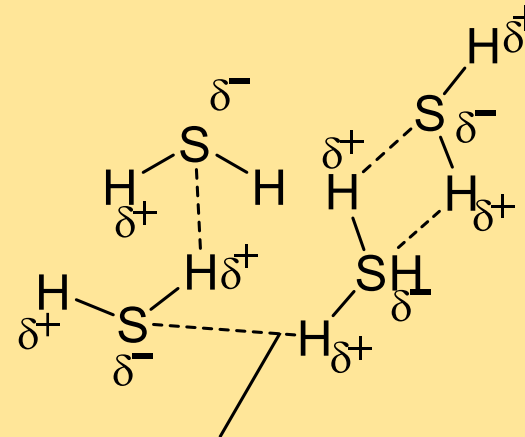
Bonding and Intermolecular Forces (*contd.*)

Other types of bonds; Intermolecular bonds which include are;

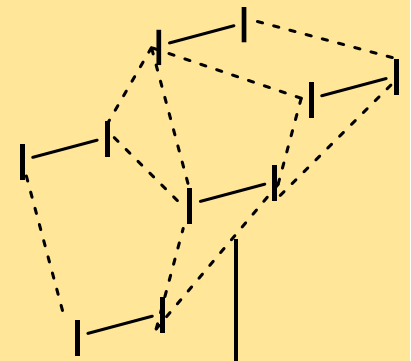
- **Hydrogen bond:** It exists between molecules containing H bonded to F, O or N.
- **Dipole-Dipole force:** It exists between heteronuclear molecules that are polar such as: H_2S , CO and SO_2 .
- **London dispersion force:** It exists between homo- and heteronuclear molecules which are non-polar. I_2 , CCl_4 , H_2 .



Hydrogen bond
 H_2O



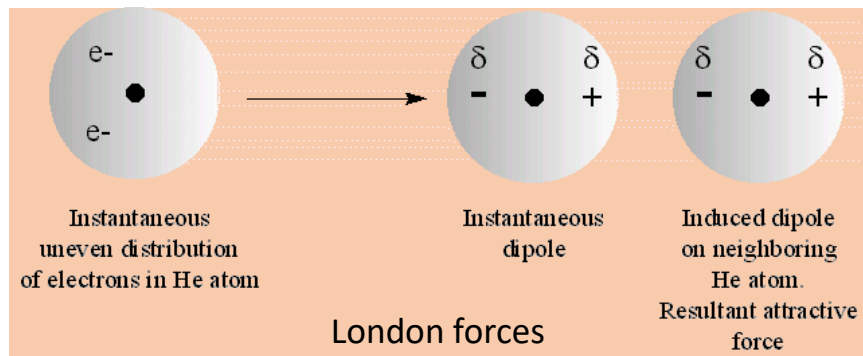
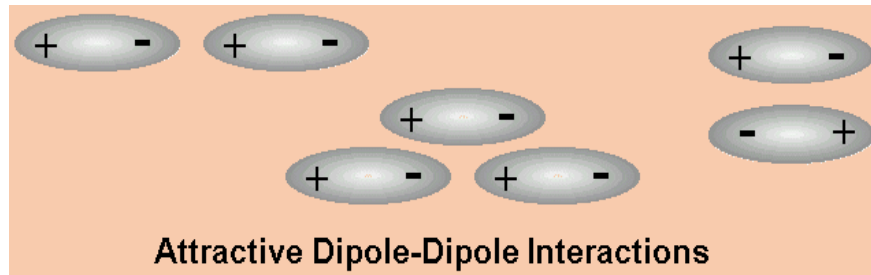
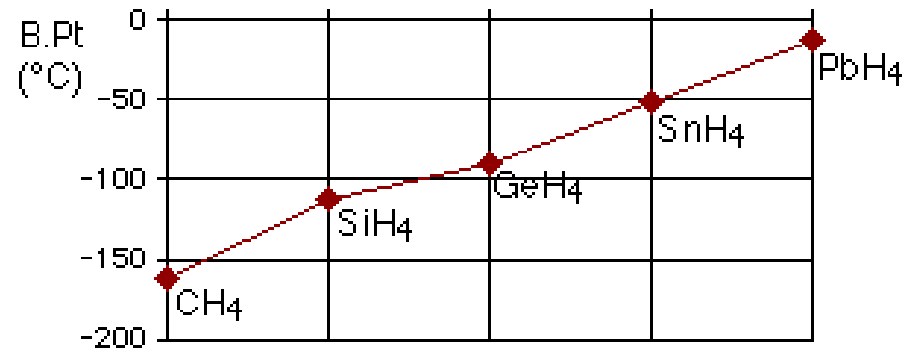
Dipole-dipole force
 H_2S



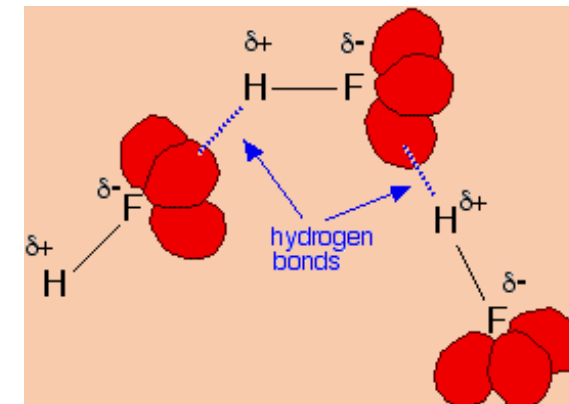
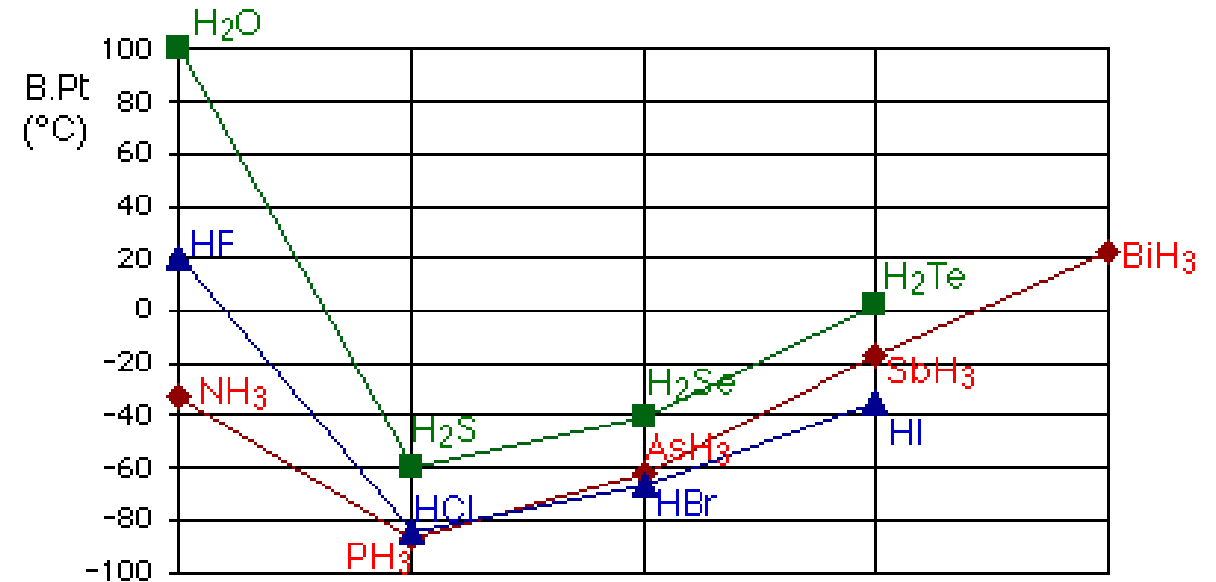
London force
 I_2

Bonding and Intermolecular Forces (*contd.*)

Non polar Hydrides of group IVA:
London forces effect on boiling point.



Effect of Hydrogen bonding in group V-VIIA



Bonding and Intermolecular Forces (*Questions*)

- Which of the following contain(s) both ionic and covalent bonds? NH_3 , NH_4Cl , MgCl_2 , CO_2 , Mg , F_2 ,
- Which of the following contain(s) only ordinary covalent bond? NH_3 , NH_4Cl , MgCl_2 , CO_2 , Mg , F_2
- Which of the following contain(s) both Dative and ordinary covalent bonds? NH_3 , NH_4Cl , MgCl_2 , CO_2 , Mg , F_2
- Which of the following contain(s) ionic bond? NH_3 , NH_4Cl , MgCl_2 , CO_2 , Mg , F_2
- Which of the following will have hydrogen bond? NH_3 , NH_4Cl , MgCl_2 , CO_2 , Mg , F_2
- Which of the following will have metallic bond? NH_3 , NH_4Cl , MgCl_2 , CO_2 , Mg , F_2
- Which of the following will have London forces? NH_3 , NH_4Cl , MgCl_2 , CO_2 , Mg , F_2
- Which of the following will have dipole-dipole forces bond? NH_3 , NH_4Cl , MgCl_2 , CO_2 , Mg , F_2

The Periodic Table and Periodicity

The Periodic Table: This consists elements arranged in order of atomic number into 18 groups and 7 periods.

- The groups are divided into main (representative) groups, transition groups
- The main groups are labelled IA-VIIIA while the transition groups are labelled IB-VIIIB.
- The elements are distributed into different blocks.
- The block of any element is determined by the location of its last valence electron when at ground state.
- The transition elements are located in d-block while the inner transition are located in f-block

Groups (IUPAC)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18

(US traditional)
IA IIA IIIB IVB VB VIB VIIIB VIIIB IB IIB IIIA IVA VA VIA VIIA VIIIA

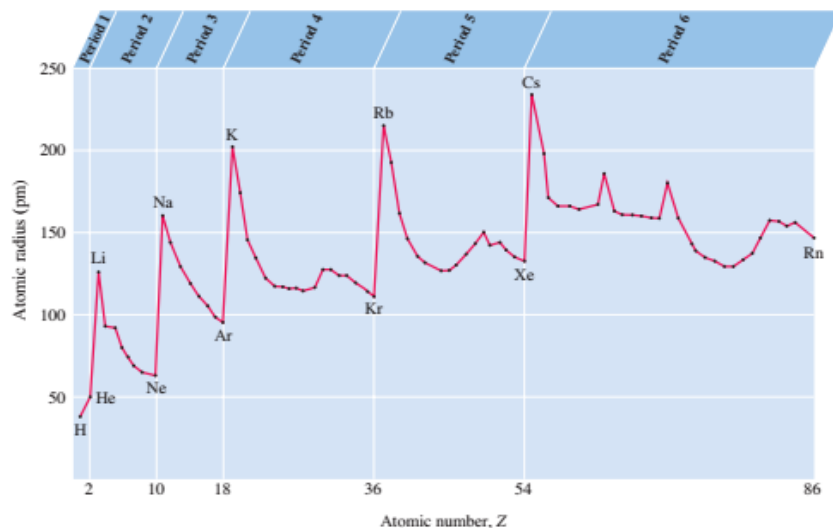
Legend:
 s block: s block
 p block: p block
 d block: d block
 f block: f block

58	59	60	61	62	63	64	65	66	67	68	69	70	71
Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
90	91	92	93	94	95	96	97	98	99	100	101	102	103
Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr

The Periodic Table and Periodicity (*contd.*)

Periodicity: This is the recurrent variation in properties of elements, when arranged in order of atomic number.

- Physical properties such as atomic radius, first ionization energy, electron affinity, ionic radius and electronegativity show variation within the groups and across the periods.
- **Atomic radii** of elements increase down the groups due to additional shell and decrease across the period as the valence shell shrinks with increase in nuclear charge.
- Atomic radius theoretically is the average distance between the center of the nucleus and the valence shell.



Nonpolar Covalent Radii (pm)

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
H 32																	He 31
Li 123	Be 89											B 82	C 77	N 75	O 73	F 71	Ne 69
Na 154	Mg 136											Al 118	Si 111	P 106	S 102	Cl 99	Ar 98
K 203	Ca 174	Sc 144	Ti 132	V 122	Cr 118	Mn 117	Fe 117	Co 116	Ni 115	Cu 117	Zn 125	Ga 126	Ge 122	As 120	Se 117	Br 114	Kr 111
Rb 216	Sr 191	Y 162	Zr 145	Nb 134	Mo 130	Tc 127	Ru 125	Rh 125	Pd 128	Ag 134	Cd 148	In 144	Sn 140	Sb 140	Te 136	I 133	Xe 126
Cs 235	Ba 198	La 169	Hf 144	Ta 134	W 130	Re 128	Os 126	Ir 127	Pt 130	Au 134	Hg 149	Tl 148	Pb 147	Bi 146	Po (146)	At (145)	Ra

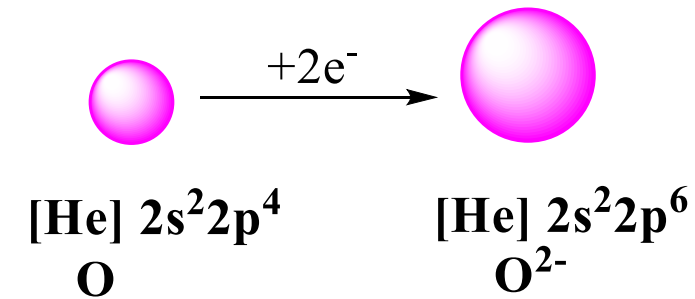
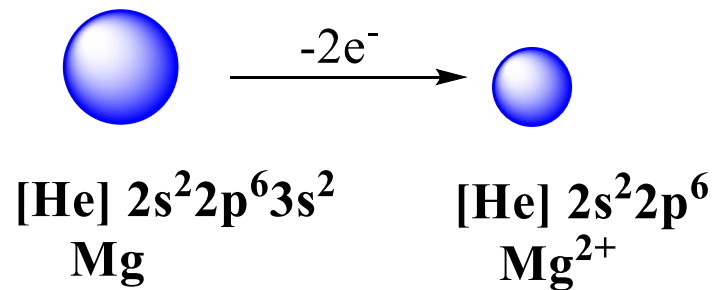
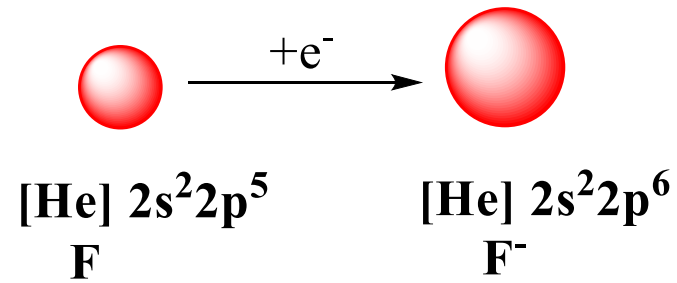
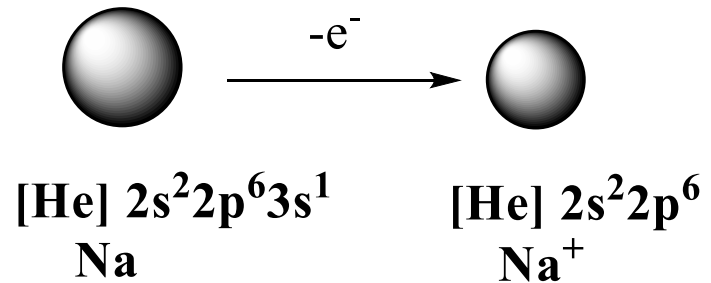
Source: R. T. Sanderson, *Inorganic Chemistry*, Reinhold, New York, 1967, p. 74; and E. C. M. Chen, J. G. Dojahn, and W. E. Wentworth, *J. Phys. Chem. A*, 1997, 101, 3088.

The Periodic Table and Periodicity (*contd.*)

- **The ionic radii of cations** (positively charged ions) are smaller than the corresponding atomic radii.
- **The ionic radii of anions** (negatively charged ions) are smaller than the corresponding atomic radii.
- Examples: Na^+ (116 pm) < Na (154 pm) Mg^{2+} (86 pm) < Mg (130 pm),

O^{2-} (126 pm) > O (73 pm)

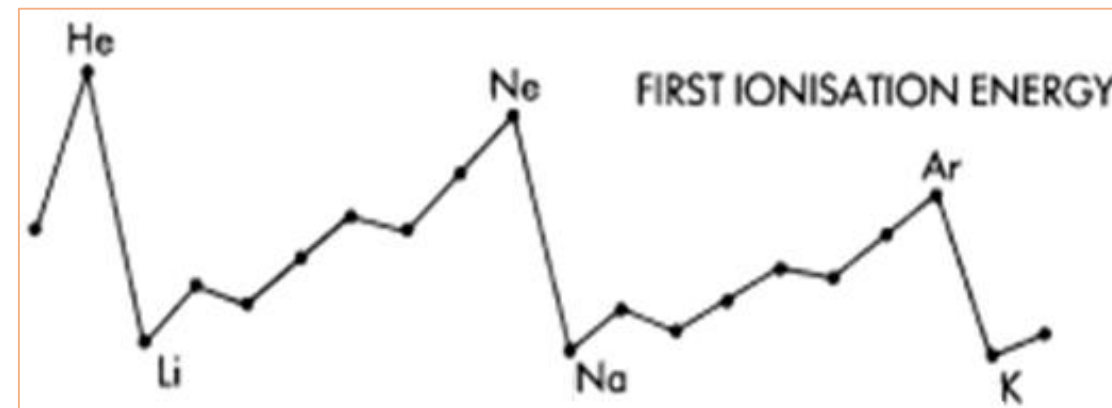
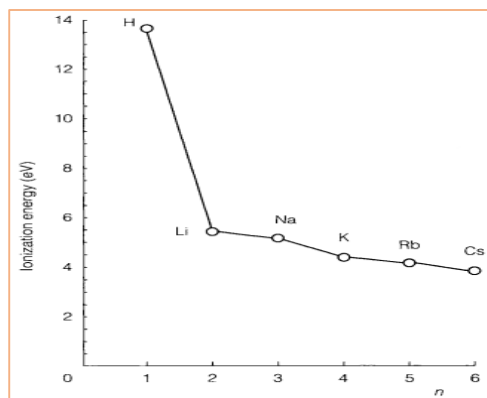
F^- (119 pm) > F (71 pm)



The Periodic Table and Periodicity (*contd.*)

- **First ionization energy** is the minimum amount of energy required to remove the most loosely bound electron from the valence shell of one mole of isolated gaseous atoms to form ions with a unit positive charge.
- $\text{Be} (1s^2 2s^2) \rightarrow \text{Be}^+ (1s^2 2s^1) + e^-$ First ionization energy
- $\text{Be}^+ (1s^2 2s^1) \rightarrow \text{Be}^{2+} (1s^2) + e^-$ Second ionization energy,
- Ionization energy decreases down the group. but increases across the period.

Element	Successive Ionization Energies (kJ/mol)			
	First	Second	Third	Fourth
Na	496	4,562	6,912	9,543
Mg	738	1,451	7,733	10,540
Al	578	1,817	2,745	11,577



The Periodic Table and Periodicity (*contd.*)

The **electron affinity** is the energy change that occurs when an electron is added to a neutral atom in the gaseous state to form a negative ion. $\text{Cl} + \text{e}^- \rightarrow \text{Cl}^-$

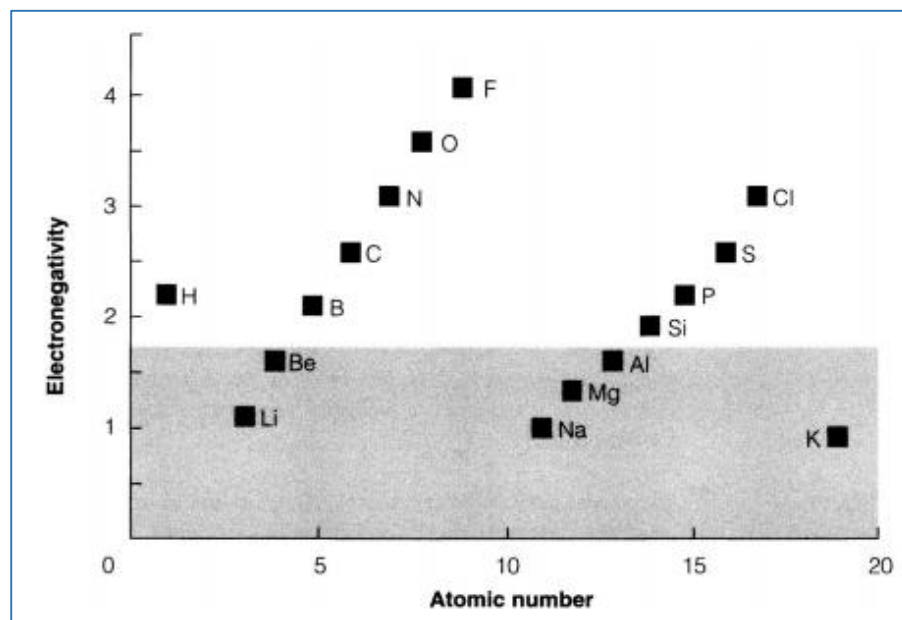
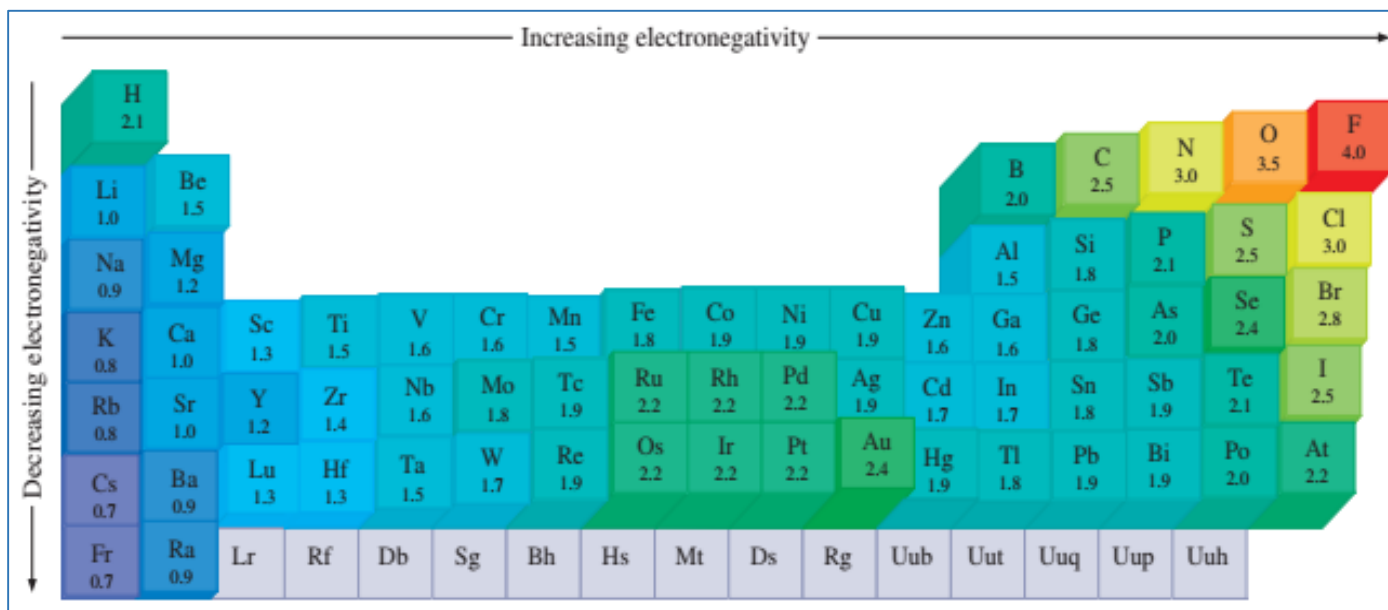
- It increases down the group but decreases across the period, though not in a regular pattern.
- The negative energy implies energy is released when electron is added.
- The higher the energy released the greater the stability of the anion formed.

1A (1)						8A (18)
H −72.8						He (0.0)
2A (2)		3A (13)	4A (14)	5A (15)	6A (16)	7A (17)
Li −59.6	Be ≤0	B −26.7	C −122	N +7	O −141	F −328
Na −52.9	Mg ≤0	Al −42.5	Si −134	P −72.0	S −200	Cl −349
K −48.4	Ca −2.37	Ga −28.9	Ge −119	As −78.2	Se −195	Br −325
Rb −46.9	Sr −5.03	In −28.9	Sn −107	Sb −103	Te −190	I −295
Cs −45.5	Ba −13.95	Tl −19.3	Pb −35.1	Bi −91.3	Po −183	At −270
						Xe (+29)
						Ar (+35)
						Kr (+39)
						Xe (+41)
						Rn (+41)

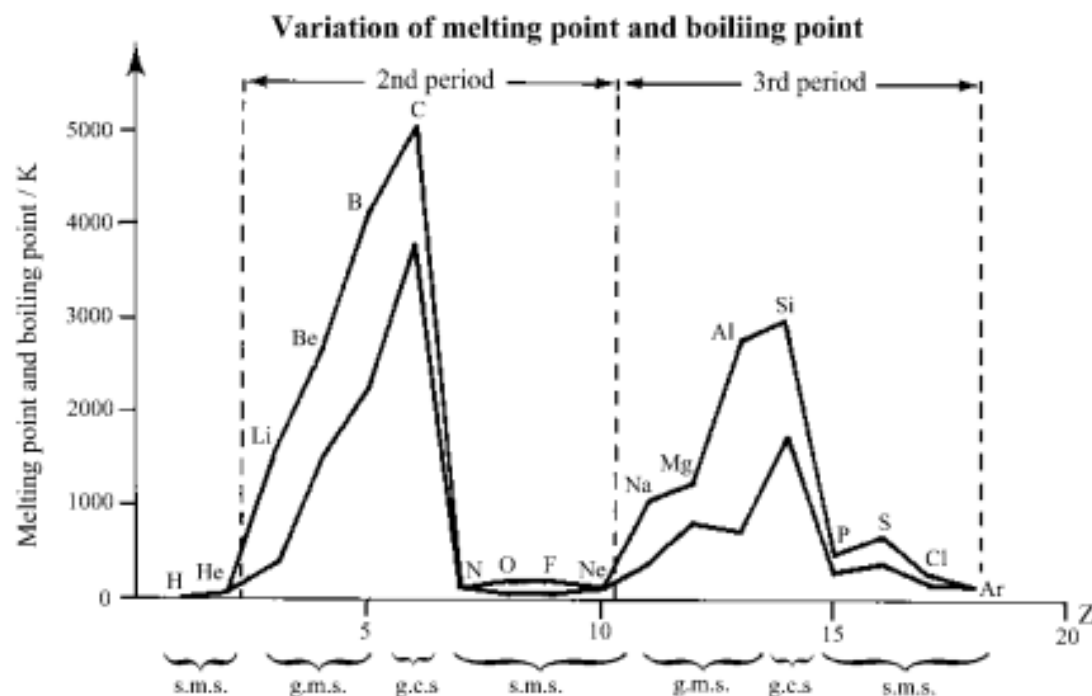
The Periodic Table and Periodicity (*contd.*)

Electronegativity is a measure of the relative ability of an atom in a molecule to attract the bonding electrons (shared pair) to itself. Electronegativity increases across the period and decreases down the group.

Pauling electronegativity is based on bond energies, using the empirical observation that bonds between atoms with a large electronegativity difference tend to be stronger than those where the difference is small.



The Periodic Table and Periodicity (*contd.*)



- Key -**
- s.m.s. : Simple molecular structure
The forces among the molecules are weak van der Waals' forces, thus the m.p. and b.p. are rather low. P₄ molecule is smaller than S₈ molecule, therefore S has a higher m.p. and b.p. than P.
 - g.m.s. : Giant metallic structure
The m.p. and b.p. of a metal is depending on the strength of the metallic bond.
 - g.c.s. : Giant covalent structure
The attraction amount the atoms are strong covalent bond, thus all giant covalent structures have very high m.p. and b.p.

simple molecular structure

giant metallic structure

giant covalent structure

The Periodic Table and Periodicity (*Questions*)

1. Which of the following has the lowest first ionization energy (a) Ca (b) Mg (c) Al (d) O
2. Which of the following has the largest ionic radius (a) Ca^{2+} (b) S^{2-} (c) Al^{3+} (d) Mg^{2+}
3. Which of the following has the smallest ionic radius (a) Ca^{2+} (b) S^{2-} (c) Al^{3+} (d) Mg^{2+}

9.20 The radii of the species S , S^+ , and S^- decrease in the following order:

- a. $S^+ > S > S^-$
b. $S^+ > S > S$
c. $S > S^- > S^+$
d. $S > S^+ > S^-$
e. $S^- > S > S^+$

9.21 Which of the following is the ground-state electron configuration of a C^{3-} ion?

- a. $1s^2 2s^2 2p^4$
b. $[\text{He}] 2s^2 2p^6$
c. $[\text{He}] 2s^1$
d. $1s^2 2s^2 2p^5$
e. $1s^2 2s^2 2p^6 3s^1$

9.22 The element X below could be



- sulfur
- iodine
- aluminum
- phosphorus
- silicon

Select the better choice in each of the following, and explain your selection briefly.

- Higher ionization energy: Ca or Ga
- Higher ionization energy: Mg or Ca
- Higher electron affinity: Si or P
- More likely configuration for Mn^{2+} : $[\text{Ar}]4s^23d^3$ or $[\text{Ar}]3d^5$

In each of the following pairs, pick the element with the higher ionization energy and explain your choice.

- a. Fe, Ru b. P, S c. K, Br d. C, N e. Cd, In

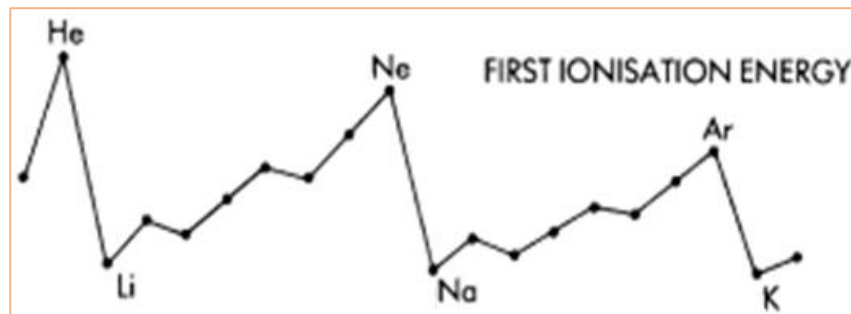
9.45 Arrange the members of each of the following pairs in order of increasing radius and explain the order:

- a. Sr, Sr^{2+} b. Br, Br^-

9.46 Arrange the members of each of the following pairs in order of increasing radius and explain the order:

- a. $\text{Te}, \text{Te}^{2-}$ b. $\text{Al}, \text{Al}^{3+}$

State the reason for the decrease in ionization energy of O with respect to N.



Descriptive Inorganic chemistry of Main groups

- The chemical and physical properties of the elements in the main groups show periodic trends.
- The elements in groups IA, IIA and IIIA are metallic while group IVA contains elements with nonmetallic, metalloid and metallic characters.
- Moving farther left to groups V-VIIA, elements show strong nonmetallic character.
- The variants of metallic–nonmetallic character are related to variations in the ionization energies the corresponding atoms.
- Those with low ionization energy have strong metallic characteristics, those with high ionization energy show stronger tendency for nonmetallic character.

				H				
	IA	IIA	IIIA	IVA	VA	VIA	VIIA	VIIIA He
2	Li	Be	B	C	N	O	F	Ne
3	Na	Mg	Al	Si	P	S	Cl	Ar
4	K	Ca	Ga	Ge	As	Se	Br	Kr
5	Rb	Sr	In	Sn	Sb	Te	I	Xe
6	Cs	Ba	Tl	Pb	Bi	Po	At	Rn
7	Fr	Ra						

■ Metal
■ Metalloid
■ Nonmetal

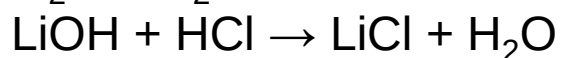
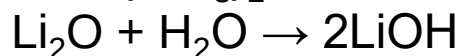
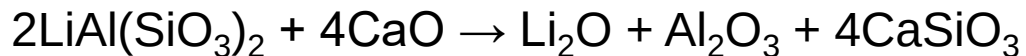
Descriptive Inorganic chemistry of Main groups (contd.)

- **Group IA** elements are also known as the alkali metals. They are very reactive and soft. They exist as compounds such as chloride, silicate, carbonate and nitrate.
- **Elements** in the group are: ${}^3\text{Li}$ ($[\text{He}] 2s^1$), ${}^{11}\text{Na}$ ($[\text{Ne}] 3s^1$), ${}^{19}\text{K}$ ($[\text{Ar}] 4s^1$), ${}^{37}\text{Rb}$ ($[\text{Kr}] 5s^1$), ${}^{55}\text{Cs}$ ($[\text{Xe}] 6s^1$) and ${}^{87}\text{Fr}$ ($[\text{Rn}] 7s^1$).
- **Sources:** The elements occur in nature as salts, examples:
 - Lithium (Li) is found Spodumene ($\text{LiAl}(\text{SiO}_3)_2$),
 - sodium (Na) is found in Rock salt (NaCl), Saltpetre (NaNO_3), Borax ($\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$) and Trona ($\text{Na}_2\text{CO}_3 \cdot \text{NaHCO}_3 \cdot 2\text{H}_2\text{O}$).
 - Potassium (K) is found in Sylvite (KCl), Sylvinite (KCl/NaCl) and Carnalite ($\text{KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$).
 - Rubidium (Rb) and Caesium (Cs) have no known suitable natural sources of commercial value.
 - Francium is radioactive in nature



Descriptive Inorganic chemistry of Main groups (contd.)

Extraction: Lithium and sodium are obtained by electrolysis of their fused halide salts.



For the extraction of sodium, the electrolyte is a mixture of 40% sodium chloride and 60% calcium chloride (as impurity) in a Down cell.



Potassium is obtained by reduction of KCl with sodium vapour.

Rubidium and Caesium are produced in a similar manner using calcium under reduced pressure. Fr is obtained by radioactivity

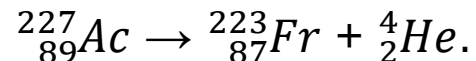
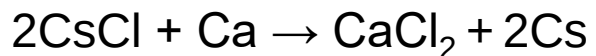
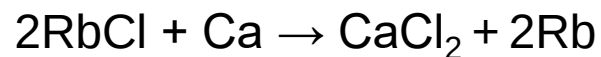
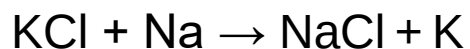


FIGURE 21.10 ▲
Cutting of sodium metal

The metal is easily cut with a sharp knife.



FIGURE 21.11 ▲
A roll of lithium metal for batteries

The metal must be handled in humidity-free rooms to prevent corrosion.

Descriptive Inorganic chemistry of Main groups (contd.)

Table 11.1 Relevant Data for Group IA, Group IIA, and Aluminum.							
Metal	Crystal structure	m.p., °C	b.p., °C	Density, g/cm ³	$-\Delta H_{\text{hydr}}$, kJ/mol	R_{atom} , pm	R_{ion} ^a , pm
Group IA							
Li	bcc	180.5	1342	0.534	515	134	68
Na	bcc	97.8	893	0.970	406	154	95
K	bcc	63.3	760	0.862	322	227	133
Rb	bcc	39	686	1.53	293	248	148
Cs	bcc	28	669	1.87	264	265	169
Group IIA							
Be	hcp	1278	2970	1.85	2487	113	31
Mg	hcp	649	1090	1.74	2003	160	65
Ca	fcc	839	1484	1.54	1657	197	99
Sr	fcc	769	1384	2.54	1524	215	113
Ba	bcc	725	1805	3.51	1360	217	135
Ra	bcc	700	1140	5	~1300	220	140
Group IIIA							
Al	fcc	660	2327	2.70	4690	143	50
^a This is the radius of the ion having the charge equal to the group number.							

Physical Properties of Group IA Metals

- They are silvery-white in appearance except caesium, which has a golden-yellow appearance.
- The atomic and ionic radii of elements in this group increases down the group.
- All the metals in the group are soft with lithium being the hardest.
- The metals have very low density
- They have the lowest first ionization energy and electronegativity
- The heat of hydration (ΔH_{hydr}) varies with size. It is the heat released following infinite dilution of mole of a gaseous ion.

Descriptive Inorganic chemistry of Main groups (contd.)

Chemical Properties of Group IA Metals

➤ **Reaction with water:** All the metals in the group react with H_2O .

➤ $2\text{M}_{(s)} + 2\text{H}_2\text{O}_{(l)} \rightarrow 2\text{MOH}_{(aq)} + \text{H}_{2(g)}$, M: Li, Na, K, Rb, Cs

➤ **Reaction with air:** The metals in this group tarnish in air due to their high reactivity.

$4\text{Li} + \text{O}_2 \rightarrow 2\text{Li}_2\text{O}$ (monoxide)

$2\text{Li} + \text{O}_2 \rightarrow \text{Li}_2\text{O}_2$ (very small peroxide)

$6\text{Li} + \text{N}_2 \rightarrow 2\text{Li}_3\text{N}$ (nitride)

$2\text{Li}_3\text{N} \rightarrow 6\text{Li} + \text{N}_2$ (decomposition)

$\text{Li}_3\text{N} + 3\text{H}_2\text{O} \rightarrow 3\text{LiOH} + \text{NH}_3$ (reaction with H_2O)

➤ Sodium forms higher proportion of peroxide than monoxide in air. K forms superoxide in addition to peroxide

Monoxide: $4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$ (oxidation state of O is -2)

$2\text{Na}_2\text{O}_2 \rightarrow 2\text{Na}_2\text{O} + \text{O}_2$

$2\text{KNO}_3 + 10\text{K} \rightarrow 6\text{K}_2\text{O} + \text{N}_2$

$4\text{KO}_2 \rightarrow 2\text{K}_2\text{O} + 3\text{O}_2$

$2\text{KNO}_3 + 10\text{K} \rightarrow 6\text{K}_2\text{O} + \text{N}_2$

Peroxide: $2\text{Na} + \text{O}_2 \rightarrow \text{Na}_2\text{O}_2$ (oxidation state of O is -1)

Superoxide: $\text{K} + \text{O}_2 \rightarrow \text{KO}_2$ (oxidation state of O is $^{-1/2}$)



FIGURE 21.13 ▲

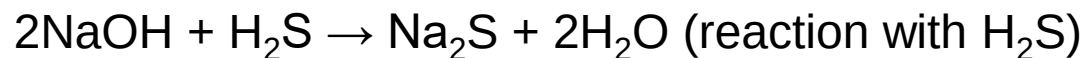
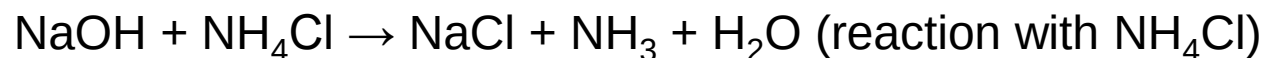
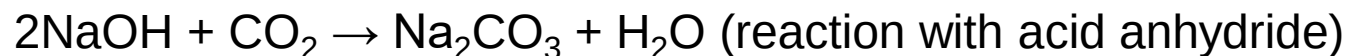
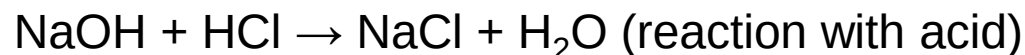
Oxygen compounds of alkali metals

Sodium peroxide, Na_2O_2 (yellowish white, left), and potassium superoxide, KO_2 (orange-yellow, right). Potassium superoxide is the principal product formed when potassium metal burns in air.

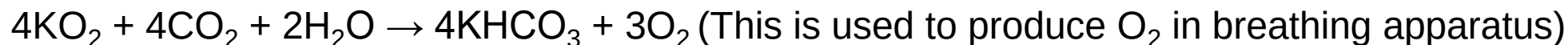
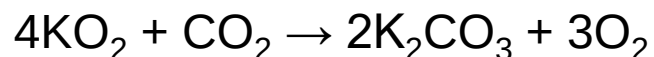
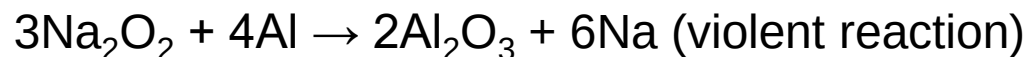
Descriptive Inorganic chemistry of Main groups (contd.)

Chemical Properties of Group IA Metals

- The hydroxides have very strong basic character with high solubility in H_2O with exception of LiOH , which is sparingly soluble.



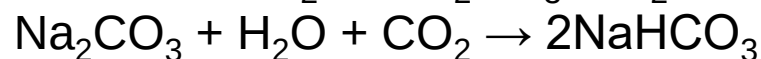
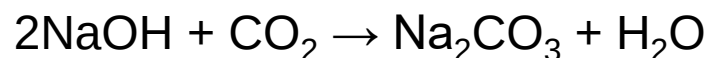
- The peroxides and superoxides also react with other compounds and elements.



Descriptive Inorganic chemistry of Main groups (contd.)

Compounds of Group IA Metals

➤ **Carbonates and bicarbonates** are formed by reaction of the metal hydroxides with CO_2 , when CO_2 is in excess, bicarbonates are formed.



Uses of NaHCO_3 : It is used as source of CO_2 in; baking powder and fire extinguisher. NaHCO_3 also has medicinal value as antacid for treatment of indigestion, heartburn and stomach upset.

Large fraction of sodium carbonate is still produced by the Solvay process.

The **Solvay process** is an industrial method for obtaining sodium carbonate from sodium chloride and limestone.

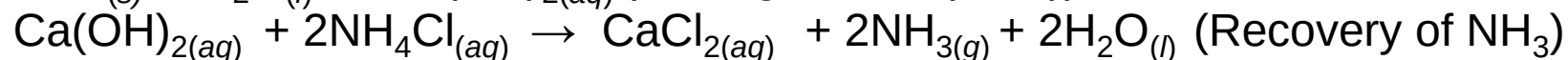
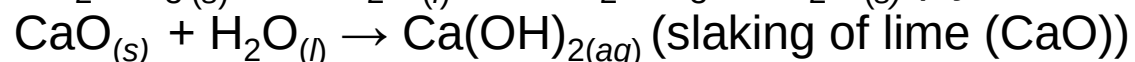
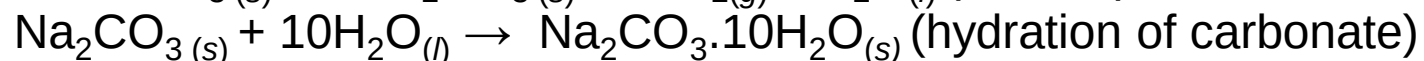
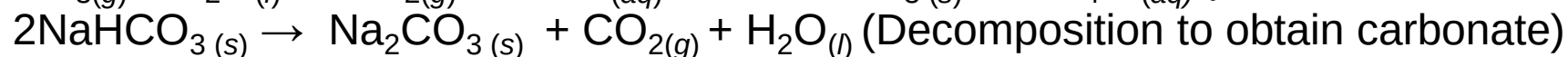


FIGURE 21.15 ▲

Sodium compounds available at the grocery store

Left to right: Salt (NaCl), baking soda (NaHCO_3), washing soda ($\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$), drain opener (NaOH), and oven cleaner (NaOH).

Descriptive Inorganic chemistry of Main groups (contd.)

Compounds of Group IA Metals

Nitrates (NO_3^-) and Nitrites (NO_2^-): The nitrates are formed by reaction of the group IA metal hydroxides or carbonates with nitric acid. The nitrites are obtained by thermal decomposition of the nitrates or by reacting metal hydroxide with NO.

$\text{NaOH} + \text{HNO}_3 \rightarrow \text{NaNO}_3 + \text{H}_2\text{O}$ (NaNO_3 is used as nitrogenous fertilizer in Chile)

$\text{K}_2\text{CO}_3 + 2\text{HNO}_3 \rightarrow 2\text{KNO}_3 + \text{H}_2\text{O} + \text{CO}_2$ (KNO_3 is used as gun powder)

$2\text{NaNO}_3 \rightarrow 2\text{NaNO}_2 + \text{O}_2$ (thermal decomposition at 500°C)

$4\text{NaNO}_3 \rightarrow 2\text{Na}_2\text{O} + 5\text{O}_2 + 2\text{N}_2$ (thermal decomposition at 800°C)

$2\text{NaNO}_3 + \text{C} \rightarrow 2\text{NaNO}_2 + \text{CO}_2$ (reduction)

$\text{KNO}_3 + \text{Zn} \rightarrow \text{KNO}_2 + \text{ZnO}$ (reduction)

$2\text{KOH} + 4\text{NO} \rightarrow 2\text{KNO}_2 + \text{N}_2\text{O} + \text{H}_2\text{O}$ (reaction with NO)

$2\text{KOH} + 6\text{NO} \rightarrow 4\text{KNO}_2 + \text{N}_2 + 2\text{H}_2\text{O}$ (reaction with NO)

Hydrides (H^-): Group IA metals react with H_2 gas to give hydrides

$2\text{M} + \text{H}_2 \rightarrow 2\text{MH}$ $\text{MH} + \text{H}_2\text{O} \rightarrow \text{MOH} + \text{H}_2$ ($\text{LiH} + \text{H}_2\text{O}$ is used for supply of H_2)

Lithium and sodium are known to form complex hydrides, which are strong reducing agents:

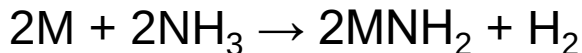
$4\text{LiH} + \text{AlCl}_3 \rightarrow \text{LiAlH}_4 + 3\text{LiCl}$ (formation of lithium aluminum tetrahydride)

$4\text{NaH} + \text{B}(\text{OCH}_3)_3 \rightarrow \text{NaBH}_4 + 3\text{NaOCH}_3$ (formation of sodium boron tetrahydride)

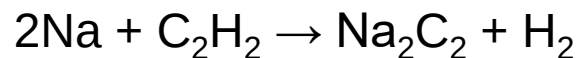
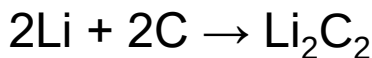
Descriptive Inorganic chemistry of Main groups (contd.)

Compounds of Group IA Metals

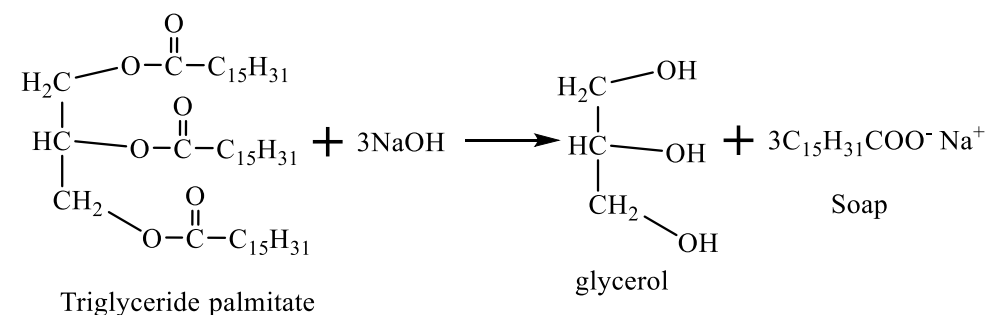
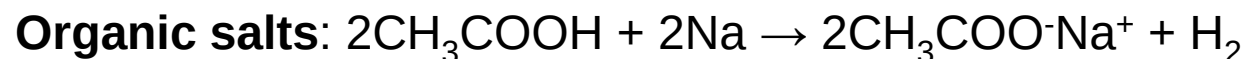
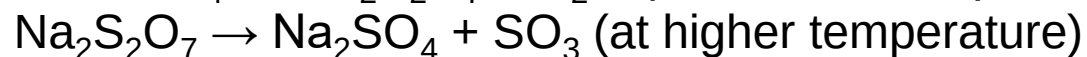
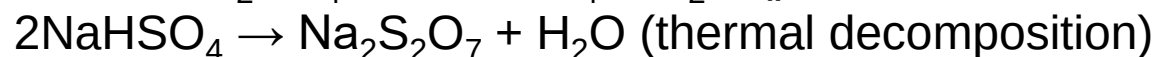
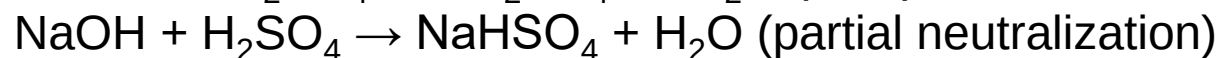
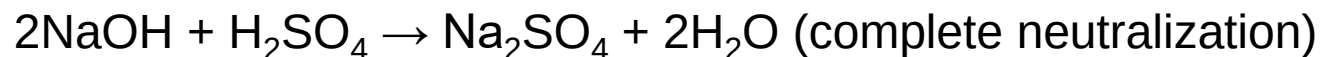
Amides (NH_2^-): Amides of group IA metals are formed by reaction of the metals with ammonia in the presence of Fe catalyst.



Carbides (C_2^{2-}): Lithium reacts directly with carbon to form carbide but others form carbides by reacting with ethyne (C_2H_2).



Sulphates (SO_4^{2-}) and bisulphates (HSO_4^-): sulphates are formed by complete neutralization of sulphuric acid with corresponding metal hydroxides, while bisulphates are formed by partial neutralization.



Descriptive Inorganic chemistry of Main groups (contd.)

Diagonal relationship of lithium with Magnesium:

Lithium exhibits anomalous behaviour comparing with other group IA metal. The properties of lithium resemble that of magnesium more than that of other group IA metals. This is an example of diagonal relationship. The anomaly is mainly due to the exceptional high charge density and polarizing power of lithium ion comparing with other group IA metal ions.

Properties	Lithium	Magnesium	Sodium
Burning in air	Formation of normal oxide	Formation of normal oxide	Formation of peroxide on top of normal oxide
Heating of carbonate	Decompose into oxide and carbon dioxide	Decompose into oxide and carbon dioxide	Stable to heat
Heating of hydroxide	Decompose into oxide and water	Decompose into oxide and water	Stable to heat
Heating of nitrate(V)	Decompose into oxide, nitrogen dioxide and oxygen	Decompose into oxide, nitrogen dioxide and oxygen	Decompose into nitrate(III) and oxygen only
Formation of hydrogencarbonate	Do not form stable solid hydrogencarbonate	Do not form stable solid hydrogencarbonate	Form stable solid hydrogencarbonate
Formation of organometallic compound	Form organometallic compound e.g. RLi	Form organometallic compound e.g. MgR_2	Do not form any stable organometallic compound at all
Solubility of salts	Insoluble phosphate, fluoride and carbonate	Insoluble phosphate, fluoride and carbonate	Soluble phosphate, fluoride and carbonate
Stability of compound	Form stable nitride	Form stable nitride	Do not form stable nitride,

Descriptive Inorganic chemistry of Main groups (contd.)

- **Group IIA** metals are also known as the alkaline earth metals. They are very reactive but less reactive than the group IA metals.
- **Elements** in the group are: ${}_4\text{Be}$ ($[\text{He}] 2s^2$), ${}_{12}\text{Mg}$ ($[\text{Ne}] 3s^2$), ${}_{20}\text{Ca}$ ($[\text{Ar}] 4s^2$), ${}_{38}\text{Sr}$ ($[\text{Kr}] 5s^2$), ${}_{56}\text{Ba}$ ($[\text{Xe}] 6s^2$) and ${}_{88}\text{Ra}$ ($[\text{Rn}] 7s^2$).
- **Sources:** The elements occur in nature as salts, examples:
 - Beryllium: beryl ($\text{Be}_3\text{Al}_2\text{Si}_6\text{O}_{18}$) and phenacite (Be_2SiO_4)
 - Magnesium: dolomite ($\text{MgCO}_3 \cdot \text{CaCO}_3$), Magnesite (MgCO_3) and talc ($\text{Mg}_3(\text{OH})_2\text{Si}_4\text{O}_{10}$)
 - Calcium is found in limestone (CaCO_3) and fluorite (CaF_2)
 - Strontium is found in strontianite (SrCO_3) and celestite (SrSO_4)
 - Barium is found in barite BaSO_4 ;
 - Radium is radioactive in nature, it is found in small quantity (1g/tonne) in uranium mineral c



FIGURE 21.17 ▲

Magnesium and barium metals

Magnesium metal turnings are in the beaker (left). Barium metal (right) is much more reactive than magnesium and must be stored in a bottle of kerosene to exclude moisture and oxygen, with which barium reacts.



beryl



limestone



dolomite

Descriptive Inorganic chemistry of Main groups (contd.)

Extraction:

- Beryllium is extracted at 350°C by electrolysis of beryllium chloride (BeCl_2), which is produced from beryl. It can also be obtained by reduction of beryllium fluoride (BeF_2) using magnesium.
$$\text{BeF}_2 + \text{Mg} \rightarrow \text{Be} + \text{MgF}_2$$
- Magnesium can be extracted by several methods; the dominant process is the electrolysis of a fused halide mixture of $\text{MgCl}_2 + \text{CaCl}_2 + \text{NaCl}$.
- From Sea water rich in Mg^{2+} and Ca^{2+} ions:
$$\text{Ca(OH)}_{2(\text{aq})} + \text{Mg}^{2+}_{(\text{aq})} \rightarrow \text{Mg(OH)}_{2(\text{s})} + \text{Ca}^{2+}_{(\text{aq})} \quad \text{(Precipitation)}$$
$$\text{Mg(OH)}_2 + 2\text{HCl} \rightarrow \text{MgCl}_2 + 2\text{H}_2\text{O} \quad \text{(Neutralization)}$$

Electrolysis

$$\text{Mg}^{2+} + 2\text{e}^- \rightarrow \text{Mg} \text{ (at the iron cathode)} \quad 2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^- \text{ (at the graphite anode)}$$
- Reduction
$$\text{MgO} + \text{C} \rightarrow \text{Mg} + \text{CO} \text{ (at } 200^\circ\text{C)}$$
$$\text{CaCO}_3 \cdot \text{MgCO}_3 \rightarrow \text{CaO} \cdot \text{MgO} + 2\text{CO}_2 \text{ (decomposition)} \quad 2\text{CaO} \cdot \text{MgO} + \text{Si} \rightarrow 2\text{Mg} + \text{Ca}_2\text{SiO}_4$$
- Calcium, strontium and barium are extracted by electrolysis of their respective halides

Descriptive Inorganic chemistry of Main groups (contd.)

Table 11.1 Relevant Data for Group IA, Group IIA, and Aluminum.							
Metal	Crystal structure	m.p., °C	b.p., °C	Density, g/cm ³	$-\Delta H_{hyd}^a$ kJ/mol	R_{atom} pm	R_{ion}^a pm
Group IA							
Li	bcc	180.5	1342	0.534	515	134	68
Na	bcc	97.8	893	0.970	406	154	95
K	bcc	63.3	760	0.862	322	227	133
Rb	bcc	39	686	1.53	293	248	148
Cs	bcc	28	669	1.87	264	265	169
Group IIA							
Be	hcp	1278	2970	1.85	2487	113	31
Mg	hcp	649	1090	1.74	2003	160	65
Ca	fcc	839	1484	1.54	1657	197	99
Sr	fcc	769	1384	2.54	1524	215	113
Ba	bcc	725	1805	3.51	1360	217	135
Ra	bcc	700	1140	5	~1300	220	140
Group IIIA							
Al	fcc	660	2327	2.70	4690	143	50
^a This is the radius of the ion having the charge equal to the group number.							

Physical Properties of Group IIA Metals

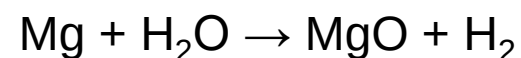
- They are mostly silvery in appearance.
- The atomic and ionic radii of metals in this group are large but smaller than those of corresponding group IA metal.
- They are harder than those in group IA.
- The metals have higher density relative to Group IA
- They have higher first ionization energy relative to Group IA

Descriptive Inorganic chemistry of Main groups (contd.)

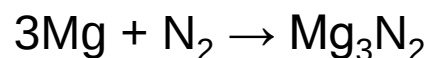
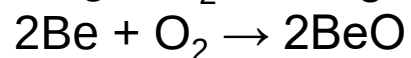
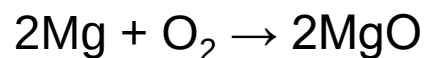
Chemical Properties of Group IIA Metals

➤ **Reaction with water:** Beryllium does not react with water but other metals in the group do.

➤ $M + 2H_2O \rightarrow M(OH)_2 + H_2$ (M: Mg, Ca, Sr and Ba)



➤ **Reaction with air:** The metals in this group burn in air to produce a mixture of oxide and nitrides.



All the oxides except beryllium oxide react with water to give hydroxides. Beryllium oxide is insoluble in water.

➤ Sodium forms higher proportion of peroxide than monoxide in air. K forms superoxide in addition to peroxide.

➤ $MO + H_2O \rightarrow M(OH)_2$ (M: Mg, Ca, Sr, Ba)

➤ Beryllium oxide is amphoteric but other group IIA oxides are basic.

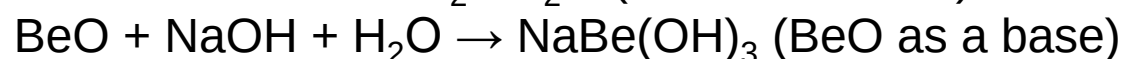
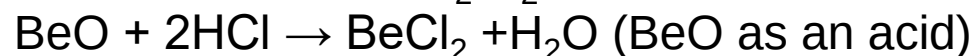
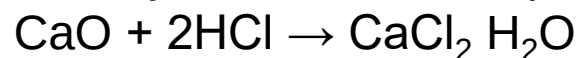
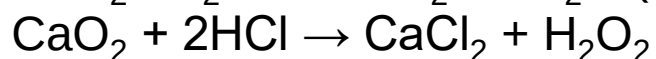
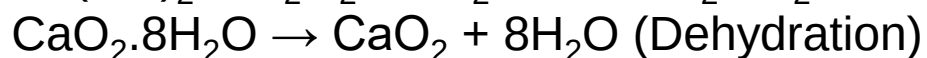
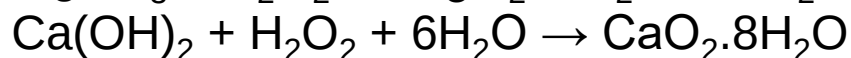


FIGURE 21.20 ▲
Milk of magnesia
Milk of magnesia is a suspension of $Mg(OH)_2$ in water. It is used as an antacid and a laxative.

Descriptive Inorganic chemistry of Main groups (contd.)

Compounds of Group IIA Metals

- **Peroxides:** BeO_2 is unknown



- **Sulphides:** $8\text{M} + \text{S}_8 \rightarrow 8\text{MS}$

- **Carbonates and bicarbonates:**

$$\text{Ca(NO}_3)_2(\text{aq}) + \text{Na}_2\text{CO}_3(\text{aq}) \rightarrow \text{CaCO}_3(\text{s}) + 2\text{NaNO}_3(\text{aq})$$

$$\text{Mg(NO}_3)_2(\text{aq}) + 2\text{NaHCO}_3(\text{aq}) \rightarrow \text{MgCO}_3(\text{s}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l}) + 2\text{NaNO}_3(\text{aq})$$

$$\text{CaCO}_3 + \text{H}_2\text{O} + \text{CO}_2 \rightarrow \text{Ca(HCO}_3)_2$$

- **Sulphates:**

$$\text{CaSO}_4 \cdot 2\text{H}_2\text{O} \rightarrow \text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O} \text{ (at } 150^\circ\text{C) (Plaster of Paris)}$$

$$\text{MSO}_4 \rightarrow \text{MO} + \text{SO}_3 \quad \text{MSO}_4 + 4\text{C} \rightarrow \text{MS} + 4\text{CO}$$

- **Nitrates:** $\text{MgCO}_3 + 2\text{HNO}_3 \rightarrow \text{Mg(NO}_3)_2 + \text{CO}_2 + \text{H}_2\text{O}$
- Hydrides:** $\text{M} + \text{H}_2 \rightarrow \text{MH}_2$ (M: Mg, Ca, Sr, Ba)
- $$\text{BeCl}_2 + 2\text{LiBH}_4 \rightarrow \text{BeB}_2\text{H}_8 + 2\text{LiCl}$$
- $$\text{BeB}_2\text{H}_8 + 2\text{PPh}_3 \rightarrow \text{BeH}_2 + 2\text{Ph}_3\text{PBH}_3$$



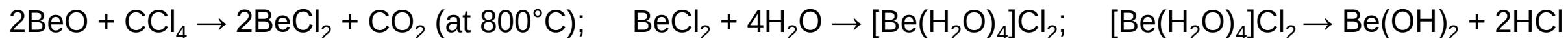
FIGURE 21.22 ▲

Some antacids contain calcium carbonate
These antacid preparations consist of purified calcium carbonate with flavorings and a binder. Such antacids are sometimes prescribed as a nutrient calcium supplement.

Descriptive Inorganic chemistry of Main groups (contd.)

Compounds of Group IIA Metals

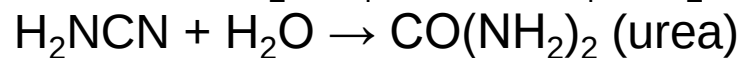
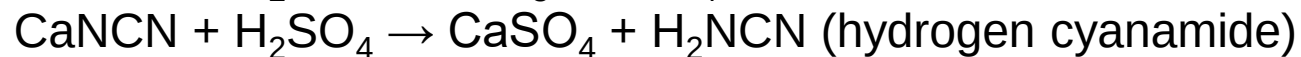
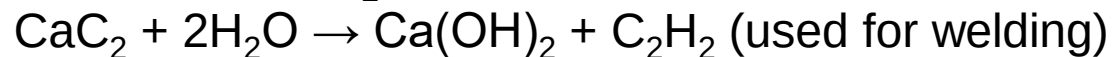
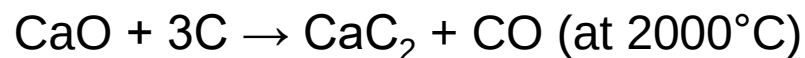
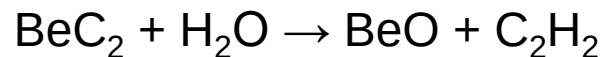
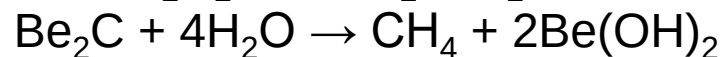
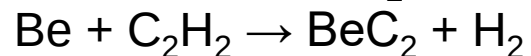
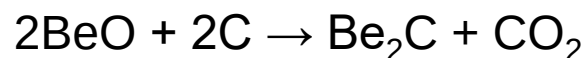
➤ **Halides:** $M + Cl_2 \rightarrow MCl_2$ (M: Be, Mg, Ca, Sr, Ba) $MCO_3 + HCl \rightarrow MCl_2 + H_2O + CO_2$ (M: Be, Mg, Ca, Sr, Ba)



➤ **Nitrides:** $3M + N_2 \rightarrow M_3N_2$



Carbides:



Organic compounds of group IIA metals



Descriptive Inorganic chemistry of Main groups (contd.)

- **Group IIIA** elements occupy the first p-block of the Periodic Table because their last valence electron is located in p-orbital ($ns^2 np^1$).
- **Elements** in the group are: ${}_5\text{B}$ ($[\text{He}] 2s^2 2p^1$), ${}_{13}\text{Al}$ ($[\text{Ne}] 3s^2 3p^1$), ${}_{31}\text{Ga}$ ($[\text{Ar}] 4s^2 3d^{10} 4p^1$), ${}_{49}\text{In}$ ($[\text{Kr}] 5s^2 4d^{10} 5p^1$) and ${}_{81}\text{Tl}$ ($[\text{Xe}] 6s^2 4f^{14} 5d^{10} 6p^1$).
- **Sources:** Boron: borax ($\text{Na}_2[\text{B}_4\text{O}_5(\text{OH})_4] \cdot 8\text{H}_2\text{O}$) and kernite ($\text{Na}_2[\text{B}_4\text{O}_5(\text{OH})_4] \cdot 2\text{H}_2\text{O}$)
Aluminum is found in bauxite ($\text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}$) and cryolite ($\text{Na}_3[\text{AlF}_6]$)
Gallium, Indium and Thallium exist as sulphides in nature.



FIGURE 21.27 ▲
Heterogeneous catalysts
Many heterogeneous catalysts use aluminum oxide as a carrier, or support.



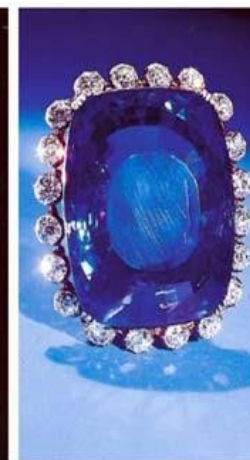
FIGURE 21.26 ▲
Protective oxide coating on aluminum
In air, aluminum metal forms an adherent coating of aluminum oxide that protects the metal from further oxidation. Note the shiny surface of the foil (left). When a similar sheet of aluminum (right) is coated with mercury, however, needle-like crystals of aluminum oxide form at the aluminum-mercury surface. Instead of adhering to the metal, the aluminum oxide flakes away from the metal, and the aluminum continuously oxidizes in air.



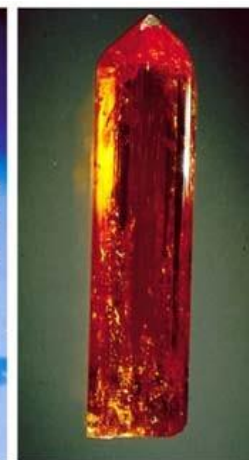
FIGURE 21.25 ▲
Aluminum products
Common items made from aluminum metal.



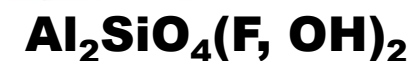
Ruby



Sapphire



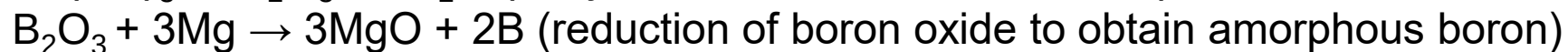
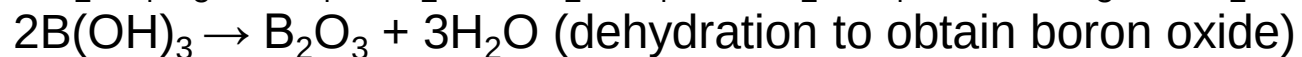
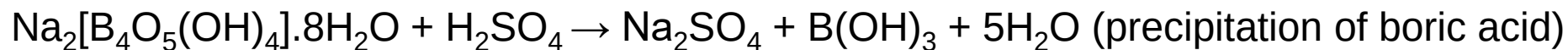
Topaz



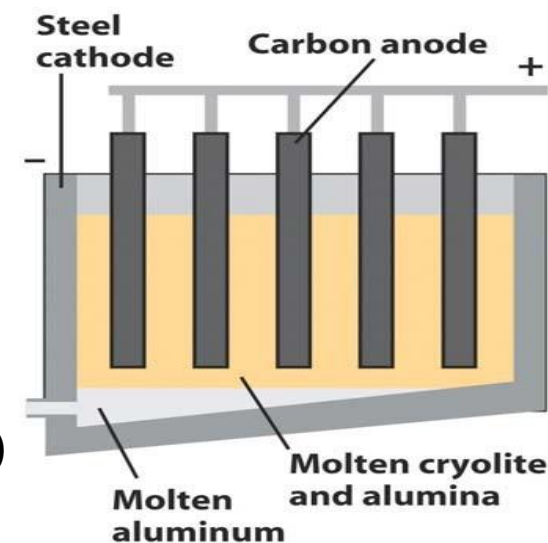
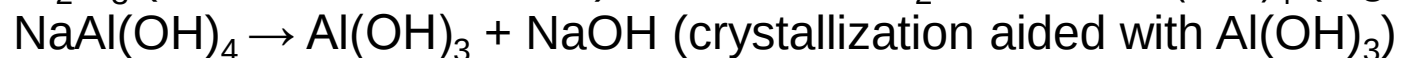
Descriptive Inorganic chemistry of Main groups (contd.)

Extraction:

Boron is extracted from borax in amorphous form.



Aluminum is extracted from bauxite:



Descriptive Inorganic chemistry of Main groups (contd.)

TABLE 14.5 Group 13/III Elements

Valence configuration: ns^2np^1

Z	Name	Symbol	Molar mass ($\text{g}\cdot\text{mol}^{-1}$)	Melting point ($^{\circ}\text{C}$)	Boiling point ($^{\circ}\text{C}$)	Density ($\text{g}\cdot\text{cm}^{-3}$)	Normal form*
5	boron	B	10.81	2300	3931	2.47	powdery brown metalloid
13	aluminum	Al	26.98	660	2467	2.70	silver metal
31	gallium	Ga	69.72	30	2403	5.91	silver metal
49	indium	In	114.82	156	2080	7.29	silver-white metal
81	thallium	Tl	204.38	304	1457	11.87	soft metal

*Normal form means the state and appearance of the element at 25°C and 1 atm.



FIGURE 8.19 ▲

Gallium

This metal melts from the heat of a hand.

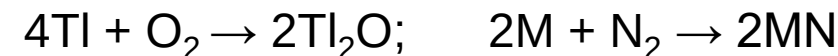
Physical Properties of Group IIIA Metals

- The metals are silvery-white in appearance except boron
- High accumulated 1st, 2nd and 3rd ionization energies give room for covalency,
- They are harder than those in group IIA.
- The tendency for the formation of +3 oxidation state decrease down the group while the tendency for +1 oxidation state increases down the group
- They have irregular melting points with boron having the highest melting and boiling points. Gallium has the lowest melting point.

Descriptive Inorganic chemistry of Main groups (contd.)

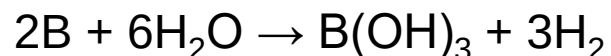
Chemical Properties of Group IIIA Metals

Reaction with air: $4M + 3O_2 \rightarrow 2M_2O_3$ (M: B, Al, Ga, In, Tl);
(M: B, Al)

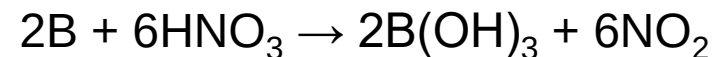
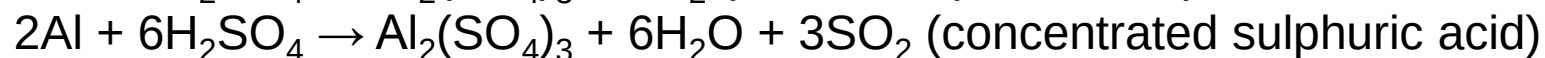
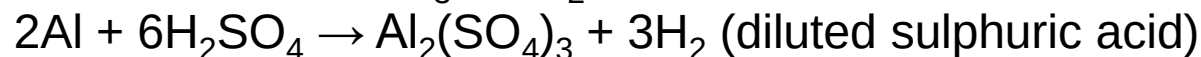
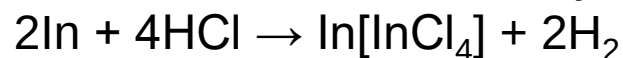
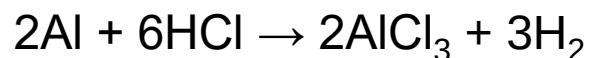


Reaction with water: B and Al (no react with water).

B reacts with steam, when it is red hot to form boric acid and hydrogen but Al will not react with steam.



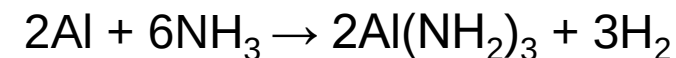
Reaction with acids: Al, Ga, In and Tl react with cold diluted HCl slowly. Boron does not react with HCl.



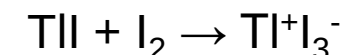
Reaction with Alkalis: $2B + 6NaOH \rightarrow 2Na_3BO_3 + 3H_2$ (fused NaOH)



Reaction with ammonia: $2B + 2NH_3 \rightarrow 2BN + 3H_2$



Reaction with halogens: $2M + 3X_2 \rightarrow 2MX_3$;



Descriptive Inorganic chemistry of Main groups (contd.)

- **Group IVA** elements show transition from nonmetallic character to metallic character down the group
- **Elements** in the group are: ${}_6\text{C}$ ($[\text{He}] 2s^2 2p^2$), ${}_{14}\text{Si}$ ($[\text{Ne}] 3s^2 3p^2$), ${}_{32}\text{Ge}$ ($[\text{Ar}] 4s^2 3d^{10} 4p^2$), ${}_{50}\text{Sn}$ ($[\text{Kr}] 5s^2 4d^{10} 5p^2$) and ${}_{82}\text{Pb}$ ($[\text{Xe}] 6s^2 4f^{14} 5d^{10} 6p^2$).
- **Sources:** Carbon: graphite and diamond; silicon: sand and quartz (SiO_2); germanium: trace in minerals of other elements; Tin: cassiterite (SnO_2); lead: galena (PbS), anglesite (PbSO_4) and cerrusite (PbCO_3).



Tin candle stand



Tin cans



Lead battery

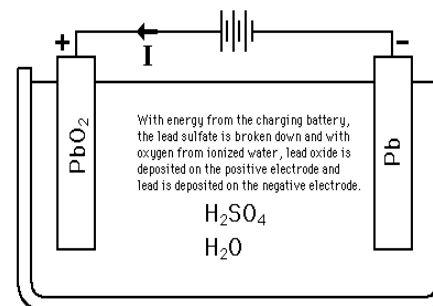


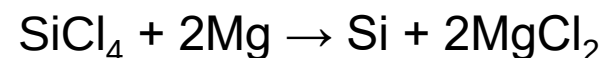
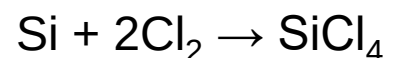
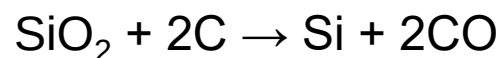
FIGURE 8.20

Oxides of some Group IVA elements
Powdered lead monoxide (yellow), lead dioxide (dark brown), tin dioxide (white), and crystalline silicon dioxide (clear quartz).

Descriptive Inorganic chemistry of Main groups (contd.)

Extraction:

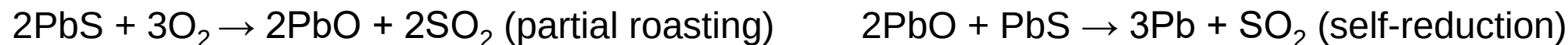
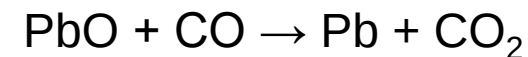
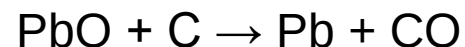
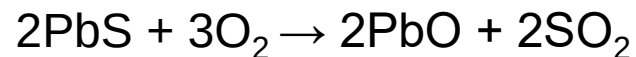
- Carbon is obtained by incomplete combustion of hydrocarbon from natural gas or oil.
- Silicon is obtained by reduction of silica (SiO_2) with high purity coke. Further purification by conversion to SiCl_4



- High purity germanium is obtained by reduction. $\text{GeO}_2 + 2\text{H}_2 \rightarrow \text{Ge} + 2\text{H}_2\text{O}$

- Tin is obtained by high temperature reduction $\text{SnO}_2 + 2\text{C} \rightarrow \text{Sn} + 2\text{CO}$ (at a temperature of 1300°C)

- Lead is extracted from galena



Descriptive Inorganic chemistry of Main groups (contd.)

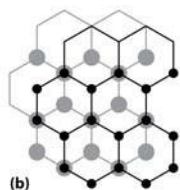
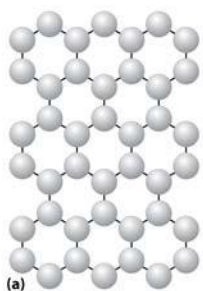
TABLE 14.6 Group 14/IV Elements

Valence configuration: ns^2np^2

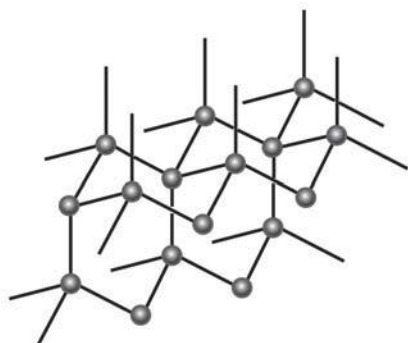
Z	Name	Symbol	Molar mass ($\text{g}\cdot\text{mol}^{-1}$)	Melting point ($^{\circ}\text{C}$)	Boiling point ($^{\circ}\text{C}$)	Density ($\text{g}\cdot\text{cm}^{-3}$)	Normal form*
6	carbon	C	12.01	3370s [†]	—	1.9–2.3 3.2–3.5	black nonmetal (graphite) transparent nonmetal (diamond) orange nonmetal (fullerite)
14	silicon	Si	28.09	1410	2620	2.33	gray metalloid
32	germanium	Ge	72.61	937	2830	5.32	gray-white metalloid
50	tin	Sn	118.71	232	2720	7.29	white lustrous metal
82	lead	Pb	207.2	328	1760	11.34	blue-white lustrous metal

*Normal form means the state and appearance of the element at 25°C and 1 atm.

[†]The symbol s denotes that the element sublimates.



**Graphite
structure**



**Diamond
structure**



**10 Buckminsterfullerene, C₆₀
Carbon Allotropes**

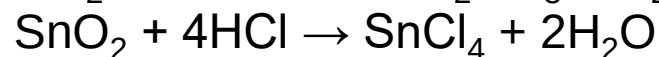
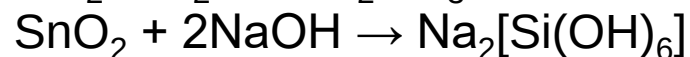
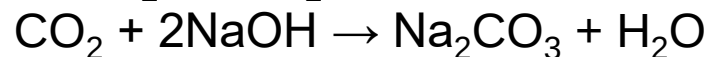
Physical Properties of Group IVA Metals

- Carbon exists as graphite (black amorphous) and diamond (colourless crystal), silicon is grey solid while other elements in the group are lustrous
- They have high melting and boiling points
- Electrical conductivity of the elements in the group shows an increase in trend down the group.
- There is tendency for the formation of +4 and +2 oxidation states.
- +4 stability decreases down the group while +2 stability increases down the group

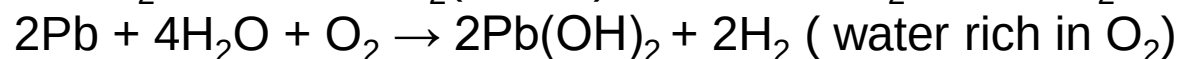
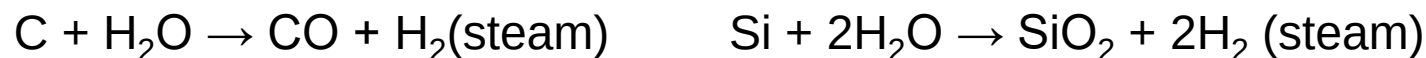
Descriptive Inorganic chemistry of Main groups (contd.)

Chemical Properties of Group IVA Metals

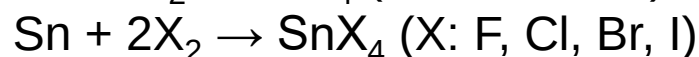
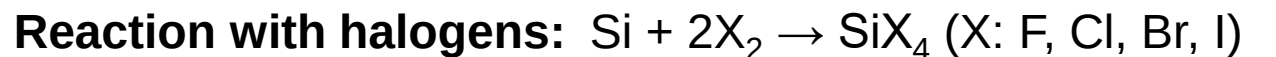
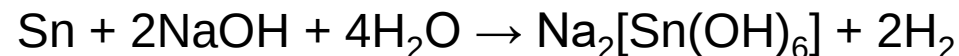
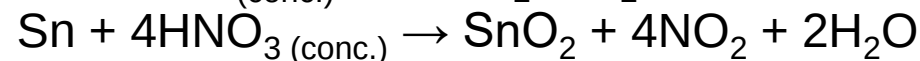
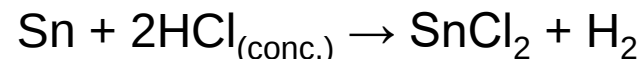
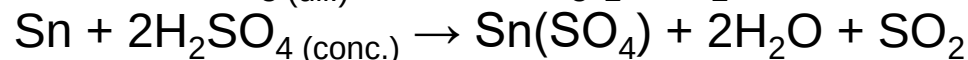
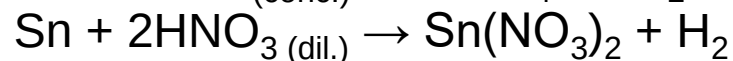
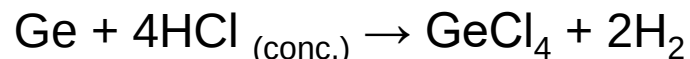
Reaction with air: Group IVA elements react with oxygen in the air at elevated temperature to form oxides; usually in +2 and +4 oxidation state depending on the supply of Oxygen.



Reaction with water:



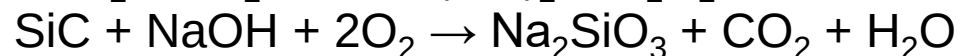
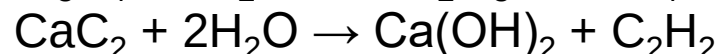
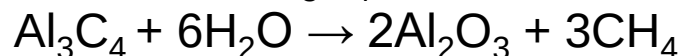
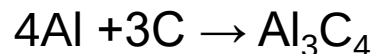
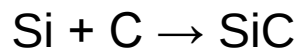
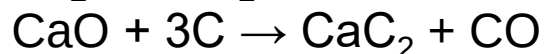
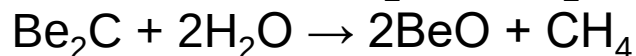
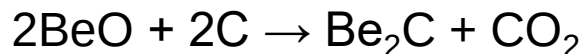
Reaction with acid:



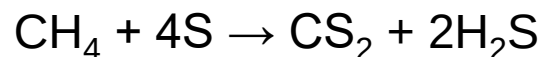
Descriptive Inorganic chemistry of Main groups (contd.)

Chemical Properties of Group IVA Metals

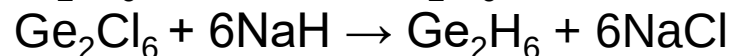
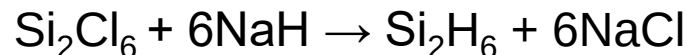
- **Carbides:** Carbon reacts with less electronegative elements to form carbides.



- **Carbon disulphide:** $\text{C} + \text{S}_2 \rightarrow \text{CS}_2$



- **Hydrides:** group IVA elements are known to form large number of hydrides.



- **Differences between carbon and other elements in group IVA**

- Carbon is the nonmetal in the group with the highest ionization energy.
- It has strong preference for the formation of covalent compounds.
- Carbon is known to form multiple bonds with itself and other electronegative elements.
- Examples include; $\text{C}=\text{C}$, $\text{C}\equiv\text{C}$, $\text{C}=\text{O}$, $\text{C}\equiv\text{N}$ and $\text{C}=\text{S}$.
- Carbon also has unique ability to catenate, which results in formation of long and branched chains as well as cyclic compounds. Silicon and germanium show catenation to a limited extent.

Descriptive Inorganic chemistry of Main groups (questions)

1. Describe the major trends that emerge when atomic radii are plotted against atomic number. Describe the trends observed when first ionization energies are plotted against atomic number.
2. What atom has the smallest radius among the alkaline earth elements?
3. What main group in the periodic table has elements with the most negative electron affinities for each period?
4. The ions Na^+ and Mg^{2+} occur in chemical compounds, but the ions Na^{2+} and Mg^{3+} do not. Explain.
5. Describe the major trends in metallic character observed in the periodic table of the elements.
6. Distinguish between an acidic and a basic oxide. Give examples of each.
7. What is the name of the alkali metal atom with valence shell configuration $5s^1$?
8. List the elements in Groups IIIA to VIA in the same order as in the periodic table. Label each element as a metal, a metalloid, or a nonmetal. Does each column of elements display the expected trend of increasing metallic characteristics?
9. For the list of elements in group IVA, note whether the oxides of each element are acidic, basic, or amphoteric.
10. A metalloid has an acidic oxide of the formula R_2O_3 . The element has no oxide of the formula R_2O_5 . What is the name of the element?
11. Two elements are in the same group, one following the other. One is a metalloid; the other is a metal. Both form oxides of the formula RO_2 . The first is acidic; the next is amphoteric (slightly acidic and basic). Identify the two elements.
12. Two elements in Period 5 are adjacent to one another in the periodic table. The ground-state atom of one element has only s electrons in its valence shell; the other has at least one d electron in an unfilled shell. Identify the elements.
13. Give the names and formulas of two oxides of carbon
14. List the unique difference of Li among group IA metals
15. What is diagonal relationship? Where does it exist?

Descriptive Inorganic chemistry of Main groups (questions)

16. Which of the following: Li, Na, K and Cs will react with both oxygen and nitrogen in the air?
17. Which of the following: Li, Na, K and Mg will form superoxide when reacted with oxygen?
18. The carbonate of group IA metals used in treatment of water hardness _____
19. An important bicarbonate in baking powder and fire extinguisher is _____
20. Which of the following salts: NaCl, Na₂CO₃, CaCO₃ and CaSO₄ will be least stable thermally?
21. The metal hydroxide used in production of lubricating soap is _____
22. Laundry soaps are made using which of the following bases: KOH, Mg(OH)₂, LiOH and NaOH?
23. Boron shows diagonal relationship with _____
24. An important impurity in the extraction of Be from BeCl₂ via electrolysis is _____
25. In the extraction of Mg from sea water, Mg²⁺ is precipitated using _____
26. List the industrial applications of Pb, Li, NaHCO₃, KO₂ and Sn
27. State the variation in the oxides of group IA metals
28. State the difference in the oxide of group IIIA and IVA
29. What makes C unique in group IVA?
30. State the reason for the difference in electrical conductivity of graphite and diamond

Brief introduction to transition metal Chemistry

- The transition metals are elements located between the s and p blocks of the Periodic Table.
- They have electrons in the d-orbitals. Hence, they are called d-block elements.
- Each row of the d-block metals consists of ten elements.
- The name, transition metals, refers to elements with partially filled d-orbitals.
- By definition; Zn, Zn^{2+} , Cu, Cu^+ , Hg, Hg^{2+} , Cd, Cd^{2+} , Ag, Ag^+ , Au and Au^+ with filled d-orbitals (d^{10}) and Ti^{4+} , Sc^{3+} , V^{5+} , Mn^{7+} , Cr^{6+} with empty d-orbitals (d^0) are d-block members but not transition metals.
- However, the study of metals with d^{10} and d^0 configuration are considered as part of transition metal chemistry.

The elements of the first transition series with their electronic configurations and known oxidation states. (Lee, 2005)

Element	Electronic configuration	Known oxidation states
$_{21}\text{Sc}$	$[\text{Ar}] 4s^2 3d^1$	+2 to +3
$_{22}\text{Ti}$	$[\text{Ar}] 4s^2 3d^2$	+2 to +4
$_{23}\text{V}$	$[\text{Ar}] 4s^2 3d^3$	+2 to +5
$_{24}\text{Cr}$	$[\text{Ar}] 4s^1 3d^5$	+2 to +6
$_{25}\text{Mn}$	$[\text{Ar}] 4s^2 3d^5$	+2 to +7
$_{26}\text{Fe}$	$[\text{Ar}] 4s^2 3d^6$	+2, +3
$_{27}\text{Co}$	$[\text{Ar}] 4s^2 3d^7$	+2, +3
$_{28}\text{Ni}$	$[\text{Ar}] 4s^2 3d^8$	+2
$_{29}\text{Cu}$	$[\text{Ar}] 4s^1 3d^{10}$	+1, +2
$_{30}\text{Zn}$	$[\text{Ar}] 4s^2 3d^{10}$	+2

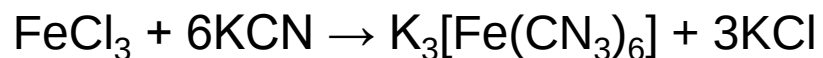
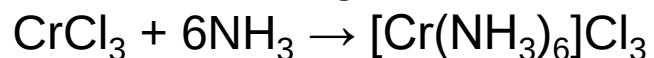
Brief introduction to transition metal Chemistry (contd.)

General properties of transition metals

- From scandium to zinc, each successive member in the first row of the transition series has one electron more than the preceding element in the 3d orbitals.
- The continuous addition of electrons to the d-orbitals in a given row for each successive metal results in decrease in atomic size across the series.
- The decrease in atomic size is not regular and it is so small that the atomic sizes of the d-block metals are very close in value. Thus, the metals share similar (not the same) physical and chemical properties.
- All the metals are solids except mercury, which is a liquid at room temperature.
- They are good conductors of heat and electricity.
- They are hard, strong and ductile with characteristic metallic lustre.
- They exist in variable oxidation states because of the ease with which the d-orbital electrons are lost. They have minimum oxidation state of +1 (Cu^+ , Ag^+ , Au^+) or +2 (Co^{2+} , Mn^{2+} , Sc^{2+}) and maximum oxidation states, which correspond to the number of electrons in their outermost orbitals (ns and (n-1)d orbitals). Exception to this occurs when the d-orbitals contain more than five electrons.

Brief introduction to transition metal Chemistry (contd.)

- The presence of the d-orbitals also allows the transition metals to form coordination compounds with coordination number greater than 4.



Atomic radii: The transition metals show decrease in atomic radii on moving from the left to the right of each row but towards the end, an increase is observed with metals having d^{10} configuration. The decrease across each row is less than expected, which is due to the ineffective shielding of the d-orbital. The decrease in atomic size can be partially attributed to;

1. The successive addition of electrons to the penultimate 3d orbitals rather than the outer 4p and 4d orbitals in the first transition series, which is a similar trend observed in the second and third rows of the transition series.
2. The poor shielding effect of the outermost shell by the d-orbitals from the nuclear attraction. On descending each group of three metals called triads, an increase is observed in atomic radii between the first and second row metals but there is little or no difference between the atomic radii of the second and third row metals. This is due to poor shielding power of the f-orbitals which contain a series of 14 elements called lanthanides.

Brief introduction to transition metal Chemistry (contd.)

3. The f-orbitals poorly shield the outermost shell from the nuclear attraction. Thus, the strong nuclear forces cause the shrinking of the atomic radii. This phenomenon is called 'lanthanide contraction'.

The atomic radii of the transition metals (10^{-10}m). (Lee, 2005)

Sc (1.44)	Ti (1.32)	V (1.22)	Cr (1.17)	Mn (1.17)	Fe (1.17)	Co (1.16)	Ni (1.15)	Cu (1.17)	Zn (1.25)
Y (1.62)	Zr (1.45)	Nb (1.34)	Mo (1.29)	Tc	Ru (1.24)	Rh (1.25)	Pd (1.28)	Ag (1.34)	Cd (1.41)
Ln (1.69)	Hf (1.44)	Ta (1.34)	W (1.30)	Re (1.28)	Os (1.26)	Ir (1.26)	Pt (1.29)	Au (1.34)	Hg (1.44)

Density: The contraction of the atomic radii of transition metals leads to decrease in volume and consequently increase in density. Transition metals are denser than s-block metals.

Brief introduction to transition metal Chemistry (contd.)

Melting and boiling points: Transition metals have very high melting and boiling points, which are often above 1000°C . This is because the d-orbitals electrons also contribute to metallic bonding. Notable exceptions are the group IIB metals, zinc, cadmium and mercury with melting point of 420°C , 321°C and -38°C respectively, where the d-orbitals electrons are all paired and not available for metallic bonding.

Ionization energy: Transition metals have high ionization energies due to their small atomic radii. Therefore, they are less reactive than s-block metals. Compound of transition metals in lower oxidation are more ionic in character than those in higher oxidation state. The higher the oxidation state the higher the covalent character. Some of the transition metals are noble in nature (not very reactive) because of high ionization energy and high energy of sublimation. The noble character is present in ruthenium, rhodium, palladium, osmium, iridium, platinum and gold. Gold and platinum are relatively stable in air, which make them very attractive as jewelries.

Colour in transition metals: The bright colours of transition metal ions are due to absorption of light energy in the visible region by the electrons in the partially filled d-orbitals. Examples are; Cu^{2+} (d^9) blue, Ti^{3+} (d^1) yellow and Fe^{2+} (d^6) green. When a transition metal ion absorbs light energy, its electrons are excited to higher energy levels. If the absorbed energy is in the visible region, the transition metal will reflect the complementary colour of the absorbed energy.

Brief introduction to transition metal Chemistry (contd.)

The colours of transition metals depend on the following:

1. The nature of the metals: Each metal ion has its unique colour. Examples; Cu^{2+} (blue), Mn^{2+} (pink), Fe^{3+} (brown) and Zn^{2+} (white).
2. The metals in different oxidation states show different colours. Examples; Fe^{2+} (green) and Fe^{3+} (brown). This is due to difference in d-electronic configuration; Fe^{2+} (d^6) and Fe^{3+} (d^5).
3. The nature of the ligands coordinated to metal ions. Examples; $[\text{Ni}(\text{NH}_3)_6]^{2+}$ (blue), $[\text{Ni}(\text{OH}_2)_6]^{2+}$ (green) and $[\text{Ni}(\text{NO}_2)_6]^{4-}$ (brownish red). This is due to effect of the ligands on the d-orbitals electrons. The ligands coordinate to the metal ions using lone pairs of electrons. The approach of the lone pair for the coordination will caused the d-orbital electron to experience a repulsive force, which depends on the influence of the ligands.
4. The colour also depends on the number of ligands coordinated and the shape of the product formed after the coordination. Examples; $[\text{Cu}(\text{OH}_2)_4]^{2+}$ (pale-blue) and $[\text{Cu}(\text{OH}_2)_6]^{2+}$ (blue).
5. The nature of the d-electronic configuration, metal ions with d^1 - d^9 configurations show different colours. Metal ions with d^0 and d^{10} are generally white solids and colourless in aqueous solution. Examples; Zn^{2+} (white) and Ti^{4+} (white). This is because they do not absorb light energy in the visible region.
6. vi. Electron charge transfer mechanism can take place in transition metal ions with d^{10} or d^0 configurations, when the ligands transfer some of their electrons to the metal empty orbitals or vice versa. Examples MnO_4^- d^0 (purple), CrO_4^{2-} d^0 (yellow) and VO_2^+ d^0 (pale yellow). The metals are coloured due to transfer of electron from the O^{2-} ions to the empty d-orbitals of the metals.

Brief introduction to transition metal Chemistry (contd.)

Magnetic properties of transition metals

- A substance that is attracted by a magnetic field is said to be magnetic. Metals contain unpaired electrons which are in continuous random motion. However, in the presence of a magnetic field, they tend to lose their randomness and become aligned in the direction of the magnetic field. This will lead to the attraction of the metal bearing the unpaired electrons in the direction of the magnetic field.
- A metal with unpaired electron(s) attracted by a magnetic field is said to be paramagnetic. A metal with no unpaired electrons are not going to be attracted by a magnetic field and are said to be diamagnetic.
- Transition metal ions with d^1 - d^9 electronic configuration are most likely to be paramagnetic based on the number of unpaired electrons in them, while those with d^0 and d^{10} are diamagnetic due to absence of unpaired electrons.
- The higher the number of unpaired electrons the greater the magnetism. Theoretically, the spin only magnetic moment ($\mu_{\text{spin only}}$) of the transition metals can be calculated using the formula:

➤ $\mu_{\text{spin only}} = \sqrt{n(n + 2)} \text{ BM}$, n = number of unpaired electrons and BM is Bohr Magnetron ($9.27 \times 10^{-24} \text{ Am}^2$)

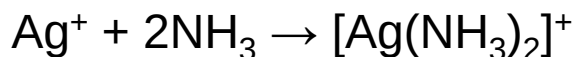
Cu^{2+}	d^9 ($n = 1$)	$\sqrt{1(1 + 2)} \text{ BM} = 1.73 \text{ BM}$
Ti^{3+}	d^1 ($n = 1$)	$\sqrt{1(1 + 2)} \text{ BM} = 1.73 \text{ BM}$
Cr^{3+}	d^3 ($n = 3$)	$\sqrt{3(3 + 2)} \text{ BM} = 3.87 \text{ BM}$

Brief introduction to transition metal Chemistry (contd.)

Formation of transition metal complexes

A transition metal complex consists of a central metal ion surrounded by a number of molecules (NH_3 , H_2O , CO) and/or negative ions (CN^- , Cl^- , O^{2-} , NH_2^-) called ligands in a coordinate bonding system. For a complex to be formed, the metal must have suitable empty orbitals to receive lone pairs of electrons from the ligands. The metal is called a Lewis acid because it is an electron pair acceptor and the ligands are called Lewis bases because they are electron pair donors.

Example:

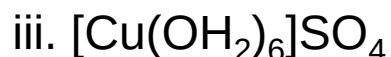
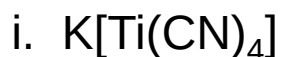


A complex is described by Alfred Werner using the concept of primary and secondary valencies. The primary valency is the oxidation state of the metal and the secondary valency is the number of ligands coordinated to the metal.

$[\text{Ag}(\text{NH}_3)_2]^+$	primary valency is +1 and secondary valency is 2
$\text{K}_4[\text{Fe}(\text{CN})_6]$	primary valency is +2 and secondary valency is 6
$[\text{Ni}(\text{CO})_4]$	primary valency is 0 and secondary valency is 4
$[\text{PtCl}_2(\text{NH}_3)_2]$	primary valency is +2 and secondary valency is 4

Questions

Determine the primary and secondary valencies of the following complexes and calculate their spin only magnetic moment.



Nuclear Chemistry

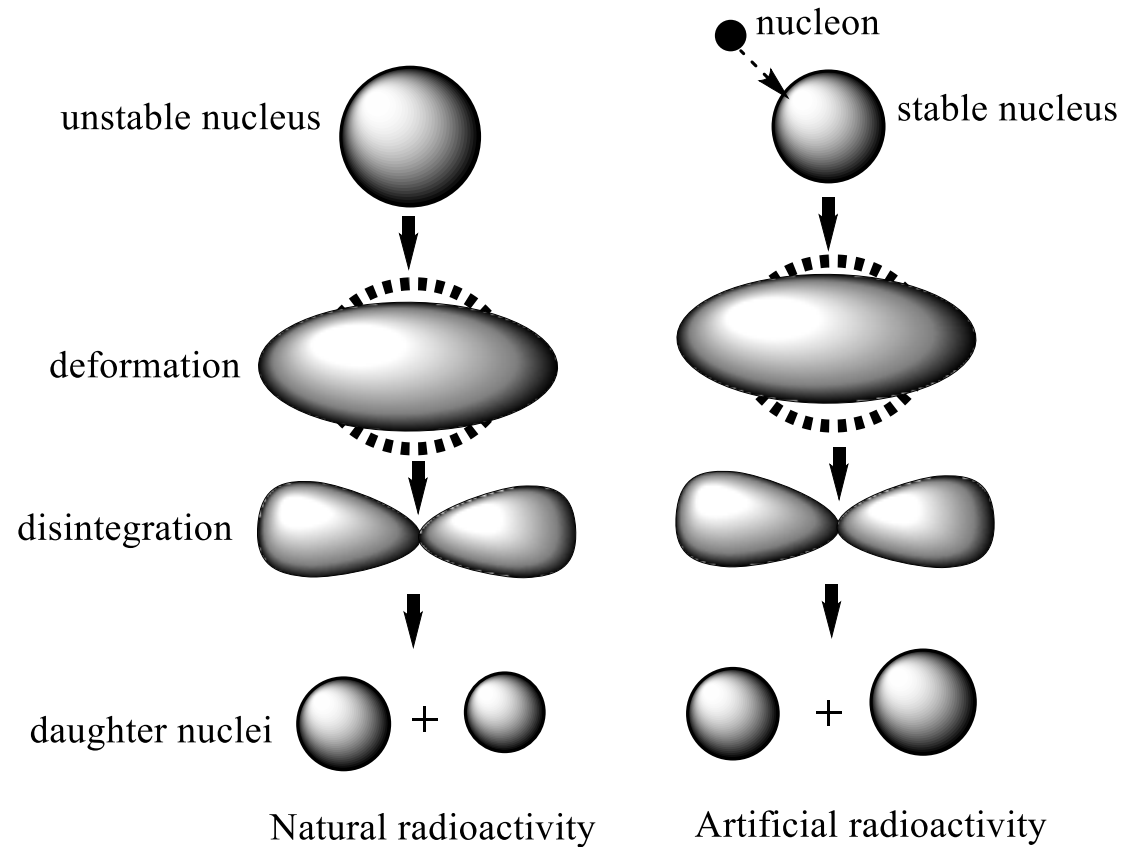
- Nuclear chemistry is the study of the nucleus, its constituents, and the behaviour of these constituents. The nucleus is the core (central part) of any given atom. According to atomic theory, an atom consists of a positively charged nucleus surrounded by cloud of negatively charged electrons revolving round the nucleus in orbits. The nucleus itself contains positively charged protons and electrically neutral particles called neutrons. Both the protons and neutrons are held together in the nucleus by short range attractive forces. The particles in the nucleus are called nucleons.

Nuclear Structure: The structure of the nucleus has been described by the liquid drop model and the shell model

- 1. The liquid drop model:** This is a model in which the nucleus is considered as a liquid drop. According to this model, increase in mass of the nucleus will lead to increase in volume. Consequently, there will be increase in surface tension of the nucleus. Therefore, deformation will become inevitable. Continuous increase in mass will cause the spherical shape of the nucleus to deform to oval shape. If the mass is further increased, with continuous vibration and rotation from the absorption of energy, the nucleus becomes unstable. Eventually, it will disintegrate to smaller unit. Similarly, a stable nucleus can disintegrate under an external force, when bombarded with a nucleon, just like a stable droplet will splash if it is impacted by a strong external force.

Nuclear Chemistry (contd.)

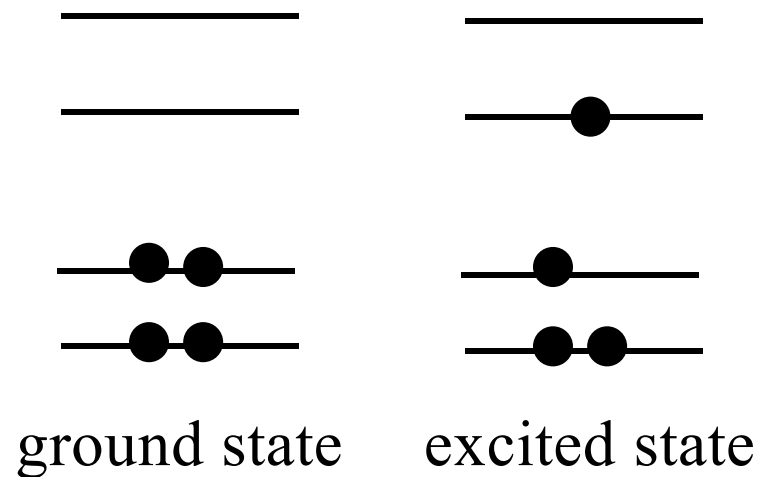
The disintegration from an unstable nucleus due to high mass is considered to be **natural radioactivity** while that resulting from impact of an external force is considered to be **artificial radioactivity**. This model was proposed in 1929 by George Gamow and adopted by Niels Bohr and John Archibald Wheeler to explain nuclear fission (disintegration of heavy nucleus) in 1939.



The illustration of natural and artificial radioactivity based on the liquid drop model.

Nuclear Chemistry (contd.)

2. **The shell model:** This model proposed that just as the electrons are located in different shells or energy levels which are described by four quantum numbers, the nucleons are also arranged in shell or energy levels in the nucleus. The nucleons are in ground state when they occupy the lowest energy level but may be excited to a different energy level under different conditions. The nucleus is stable, when the nucleons are in ground state. However, the nucleus becomes unstable if the nucleons are excited to higher energy level on absorption of energy. The excited nucleons will emit energy in the form of radiation on returning to the ground state. If the energy absorbed is too high the nucleus becomes very unstable and may disintegrate to smaller nuclei with release of various subatomic particles and energy. This process is most often spontaneous and it is called radioactivity. The shell model is more acceptable than the liquid drop model.



The illustration of the nuclear energy level based on the shell model.

Nuclear Chemistry (contd.)

Nuclear stability

The stability of any given nucleus is dependent on certain observed principle, which are:

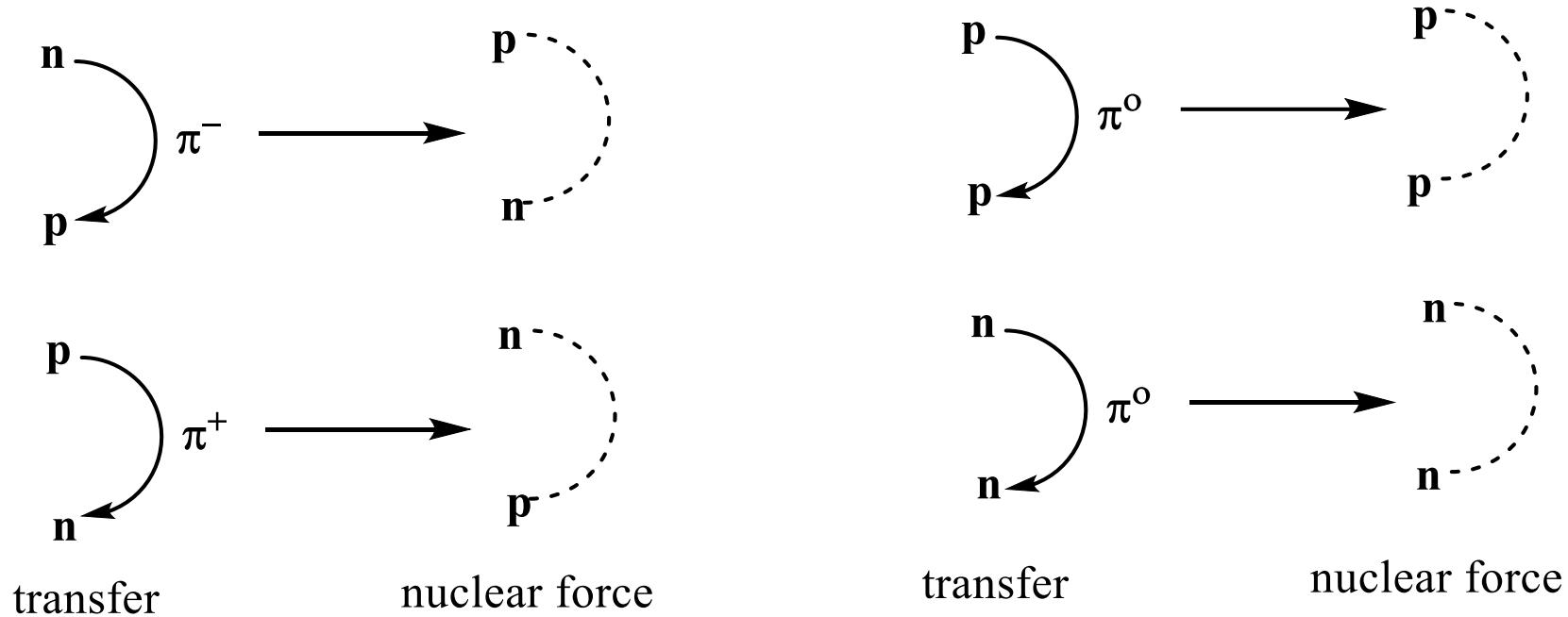
(i) Nuclear forces, (ii) Odd-even rule, (iii) Magic number and (iv) Neutron to proton ratio

Nuclear forces: The nucleus contains both protons and neutrons, which are positively charged and neutral respectively. In a nucleus with two or more protons, electrostatic repulsion will occur between the protons because like charges repel. For the repulsion not to have effect on the binding of the nucleons, there must be another force stronger than the electrostatic repulsion. This force is often referred to as nuclear force. The nuclear forces exist between proton-proton, proton-neutron and neutron-neutron. In a stable nucleus, the nuclear forces of attraction are stronger than the electrostatic repulsion between the protons. However, the nucleus becomes unstable when the nuclear forces are less than the repulsion between the protons. An unstable nucleus will disintegrate to smaller nuclei to attain stability.

The nuclear forces of attraction act within short range of $1-2.5 \times 10^{-15}$ m and it is independent of the charges on the nucleons. As two atoms may be bonded by sharing of electrons, nucleons are held together in the nucleus by sharing of a subatomic particle called meson represented with “ π ”. Nucleons are known to contain meson and it is with the exchange of mesons that nucleons can bind together. Meson may have a positive charge (π^+), a negative (π^-) or no charge (π^0).

Nuclear Chemistry (contd.)

A neutron-neutron or proton-proton attraction can take place by transfer of π^0 , while a proton-neutron attraction occurs by transfer of a π^+ or π^- . The transfer of a charged meson leads to changes in identity of the nucleons but the total number of nucleons is maintained in a stable nucleus because the changes are in equilibrium as illustrated below.



The illustration of nuclear forces of attraction between nucleons. (Lee, 2005)

Nuclear Chemistry (contd.)

Odd-even rule: This rule proposed that nuclei with even number of protons or/and neutrons are more stable than those with odd numbers. This could be related to the noble gases with completely filled shells. The following deductions are made based on odd-even rule of nuclear stability:

1. Elements with even atomic numbers (number of protons) are more stable and more abundant than nearest elements of odd atomic number, the exception in this class is hydrogen (1_1H).
2. Elements with even number of protons and neutrons are more stable than those with odd number of protons or neutrons, while those with odd numbers of protons and neutrons are rarely stable. Only four are known of such elements to be stable; 2_1H (deuterium), 6_3Li (helium), ${}^{10}_5B$ (boron) and ${}^{14}_7N$ (nitrogen).
3. Elements with even number of protons have at least 3 stable isotopes but elements with odd number of protons have at most two stable isotopes.
4. Two elements having the same mass number (sum of the numbers of protons and neutrons) but with a 1-unit difference in atomic numbers cannot both be stable. Examples: ${}^{14}_6C$ (unstable) and ${}^{14}_7N$ (stable); ${}^{24}_{11}Na$ (unstable) and ${}^{24}_{12}Mg$ (stable); ${}^{64}_{29}Cu$ (unstable) and ${}^{64}_{28}Ni$ (stable)
5. Two elements having the same mass number (sum of the numbers of protons and neutrons) and with a 2-unit difference in atomic numbers will both be stable. Examples: ${}^{64}_{28}Ni$ (stable) and ${}^{64}_{30}Zn$ (stable).

In conclusion, these rules suggest that nucleons may be paired in nucleus just as electrons are paired in shells of covalently bonded atoms; as this allows for stability. Other factors are also known to contribute to nuclear stability than pairing. A summary Table is shown below on percentage abundance of nuclei based even-odd numbers of protons and neutrons.

Nuclear Chemistry (contd.)

Odd-even rule: This rule proposed that nuclei with even number of protons or/and neutrons are more stable than those with odd numbers. This could be related to the noble gases with completely filled shells. The following deductions are made based on odd-even rule of nuclear stability:

1. Elements with even atomic numbers (number of protons) are more stable and more abundant than nearest elements of odd atomic number, the exception in this class is hydrogen (${}^1_1\text{H}$).
2. Elements with even number of protons and neutrons are more stable than those with odd number of protons or

Abundance (%)	No. of protons	No. of neutrons	Mass number
60	Even	Even	Even
40	Odd	Even	Odd
	Even	Odd	Odd
${}^2_1\text{H}$, ${}^6_3\text{Li}$, ${}^{10}_5\text{B}$ and ${}^{14}_7\text{N}$	Odd	Odd	Even

Nuclear Chemistry (contd.)

Magic number: Nuclei with certain numbers of protons and neutrons are observed to be stable. These numbers are called magic numbers, which have been associated to complete filling of the shells in nucleus just like the electronic configurations of the noble gases. The magic numbers are; 2, 8, 20, 28, 50, 82 and 126. The magic numbers stand for the number of proton or neutron and not mass number. It should be noted that the magic number is not applicable to all stable nuclei as some stable nuclei have no magic number and some unstable nuclei have magic number. Nevertheless, the concept of magic number is acceptable. Examples are shown in the Table below.

Isotope	No. of protons	No. of neutrons	Stability
${}^4_2\text{He}$	2 (magic no.)	2 (magic no.)	Stable
${}^{16}_8\text{O}$	8 (magic no.)	8 (magic no.)	Stable
${}^{23}_{11}\text{Na}$	11	12	Stable
${}^{14}_7\text{N}$	7	7	Stable
${}^{15}_8\text{O}$	8 (magic no.)	7	Unstable
${}^{14}_6\text{C}$	6	8 (magic no.)	Unstable
${}^{40}_{20}\text{Ca}$	20 (magic no.)	20 (magic no.)	Stable
${}^{64}_{28}\text{Ni}$	28 (magic no.)	36	Stable

Nuclear Chemistry (contd.)

Neutron to proton ratio (n/p): The ratio of number of neutrons to number of protons has been observed to also influence nuclear stability. In summary;

1. For stable light nuclei with atomic number up to 20, the n/p ratio should be one. Examples include; $^{16}_8\text{O}$, ^4_2He , ^6_3Li , $^{24}_{12}\text{Mg}$ and $^{14}_7\text{N}$.
2. Nuclei with high atomic number will experience greater repulsion between the protons. This is overcome by higher number of neutrons than protons to maintain stability. Stable heavy nuclei have n/p ratio of 1.6. Therefore, stability based on n/p ratio is given as $1.0 \leq n/p \leq 1.6$. Above or below this range a nucleus may become unstable and it will disintegrate through different radioactive decay modes.

Modes of radioactive decay

An unstable nucleus with n/p ratio outside the stability range will disintegrate to attain stability. The process of disintegration also called decay process is accompanied with release of radioactive particles, which can be either alpha (α), beta (β), gamma (γ) or other subatomic particles.

Alpha (α) particle: This is helium nucleus represented as ^4_2He or $^4_2\alpha$. It is positively charged with atomic number of 2 and mass number of 4. It has the lowest penetration power and the highest ionization energy. It is very slow and can be stopped by a sheet of paper.

Nuclear Chemistry (contd.)

Beta (β) particle: This is an electron and it is negatively charged. Its penetration power and speed is higher than those of alpha particle but its ionization effect is lower. It is represented as ${}_{-1}^0\beta$, which implies that it has zero mass number and charge of -1. It is faster than alpha particle and it can only be stopped by aluminum sheet.

Gamma (γ) particle: This is a radiation with zero rest mass and it is electrically neutral. It is the fastest of the three particles with the lowest ionization effect. It can only be stopped by thick block of lead.

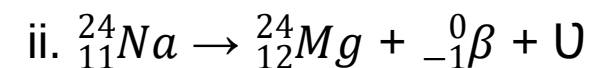
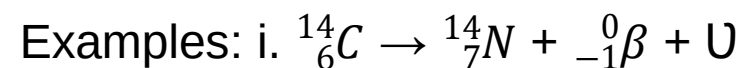
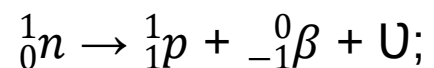
The speed (u) of a radioactive particle is inversely proportional to its rest mass (μ). Thus, the heavier the particle the slower the speed

The ionization effect of a particle is inversely proportional to the square of its speed. Thus, the faster the particle the lesser the ionization effect it will cause.

Decay modes

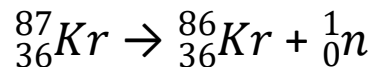
The following decay modes lead to decrease in n/p ratio:

- Beta (β) particle emission:** This process is believed to involve the transformation of a proton to a neutron, which accompany by the release of beta particle with a neutrino (U). The neutrino has zero rest mass and no charge. It is released to balance the nuclear spins. It has no effect in mass and charge balance during decay process.



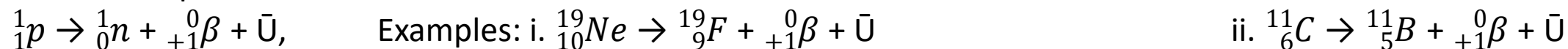
Nuclear Chemistry (contd.)

2. Neutron emission: Neutron emission can also occur to reduce the n/p ratio.

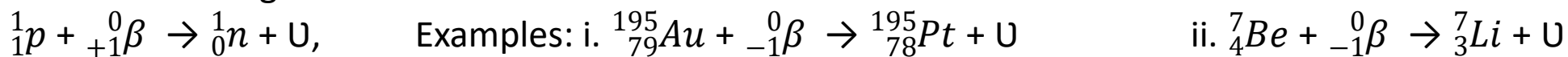


The following decay modes lead to increase in n/p ratio:

1. Positron (β^+) emission: Positron is positively charged with mass equivalent to that of electron. It is considered as the antiparticle counterpart of electron. It is also called antielectron. Positron emission involves the transformation of a proton to a neutron. This transformation is accompanied with antineutrino ($\bar{\nu}$) just to keep the nuclear spin balanced. The emission of positron results in decrease in atomic number but no increase in mass number.



2. Electron capture: This occurs when an electron in K-shell close to a nucleus is absorbed into the nucleus. This is observed, when there is insufficient energy for a positron emission to occur. The atomic number will decrease but the mass number remains unchanged.



3. Proton emission: Proton emission also leads to increase in n/p ratio but very high energy is required and it rarely occurs.

Decay mode without effect on n/p ratio:

Gamma radiation: After nuclear reactions, most of the nucleons will be in excited state. To return to ground state, the nucleons usually emit gamma radiation. Gamma radiation often accompanies most nuclear reactions.

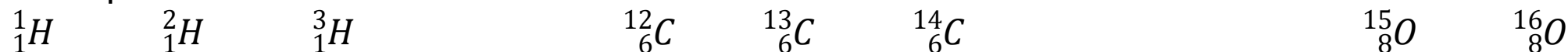


Nuclear Chemistry (contd.)

Some basic nuclear terminologies

Isotopes: These are atoms with the same atomic number but different mass numbers.

Examples:



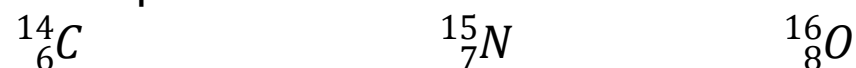
Isobars: These are atoms with the same mass number but different atomic numbers.

Examples:



Isotones: These are atoms with the same number of neutrons but different mass numbers.

Examples:



Radioactive decay constant (λ) and half-life ($t_{1/2}$)

The Half-life of a radioactive nuclei is the time required for half of a given population (or mass) of the radioactive nuclei to disintegrate. The rate of decay per unit time ($\frac{dN}{dt}$) is directly proportional to the number (N) of atoms present at a given time.

$$\frac{-dN}{dt} \propto N \quad t = \text{time taken, } N = \text{population at a given time and } \lambda = \text{decay constant}$$

$$\frac{-dN}{dt} = \lambda N$$

$$\int_{N_0}^{N_t} \frac{-dN}{N} = \int_{t_0}^t \lambda \quad N_t \text{ and } N_0 \text{ represent the population at a given time } t \text{ and at initial time } t_0$$

Nuclear Chemistry (contd.)

$$-[\ln N]_{N_0}^{N_t} = \lambda t$$

Note: $\int \frac{1}{x} dx = \ln x$

$$[\ln N_t - \ln N_0] = \lambda t$$

$$\ln\left(\frac{N_t}{N_0}\right) = -\lambda t \quad \text{or} \quad N_t = N_0 e^{-\lambda t}$$

$$N_t = \frac{1}{2}N_0 \text{ at } t = t_{1/2}$$

$$\ln\left(\frac{0.5N_0}{N_0}\right) = -\lambda t_{1/2} \quad \text{or} \quad 0.5N_0 = N_0 e^{-\lambda t_{1/2}}$$

$$t_{1/2} = \frac{-\ln 0.5}{\lambda} = \frac{0.693}{\lambda}$$

Note: $-\ln 0.5 = 0.693$

$$\lambda = \frac{0.693}{t_{1/2}}$$

Worked Example

1. If the half-life of a radioactive nucleus is 3 days what fraction of a given sample of the nucleus will remain after 12 days?

After 3 days, $\frac{1}{2}$ is left

After 6 days, $\frac{1}{2}$ ($\frac{1}{2}$) or $(\frac{1}{2})^2$ is left

After 9 days, $\frac{1}{2}$ ($\frac{1}{2}$) ($\frac{1}{2}$) or $(\frac{1}{2})^3$ is left

After 12 days, $\frac{1}{2}$ ($\frac{1}{2}$) ($\frac{1}{2}$) ($\frac{1}{2}$) or $(\frac{1}{2})^4$ is left

Therefore, fraction left after 12 days = $\frac{1}{16}$

Fraction left can also be derived using $(\frac{1}{2})^n$, where n = time given/ half-life, $n = 12/3 = 4$

Nuclear Chemistry (contd.)

ii. If the half-life of $^{222}_{86}\text{Rn}$ is 3.8 days, how long will it take for 0.05 mmol of a sample of $^{222}_{86}\text{Rn}$ to decay to 0.00625 mmol?
 $\lambda = 0.693/3.8 \text{ days} = 0.1824 \text{ day}^{-1}$

$$N_t = N_0 e^{-\lambda t}$$

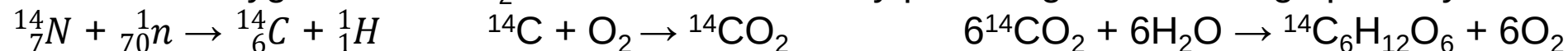
$$0.00625 = 0.05 e^{-(0.1824)t}$$

$$\frac{0.00625}{0.05} = e^{-(0.1824)t}$$

$$t = \frac{\ln 0.125}{-0.1824} = 11.4 \text{ days}$$

Carbon dating

Carbon has three known isotopes, ^{12}C , ^{13}C and ^{14}C . Carbon-14 (^{14}C) is the heaviest and it is radioactive in nature with half-life of 5730 years. Thus, it is called radiocarbon. Its long half-life makes it useful in archeological dating. Carbon-14 is formed when atmospheric nitrogen is bombarded with neutron in cosmic radiation. The carbon-14 reacts with oxygen to form CO_2 which is converted by plant to glucose through photosynthesis.



The glucose is used to build plant tissues or as a source of energy. Animal can feed on the tissues, which makes carbon-14 available in all living and dead organisms. A natural balance exists between intake of radiocarbon and loss by decay. When an organism dies, the intake stops but the decay will continue. Thus, old samples of dead plants and animals will have less carbon-14 than the corresponding new samples from living plants and animals. Comparison of the amount of carbon-14 in old and new samples of a particular plant or animal can be used to determine the age of the old sample. To determine the amount of radiocarbon left in a given sample, the sample is subjected to combustion in oxygen to generate CO_2 which is introduced into a radioactive counter to determine the amount of radiocarbon present. ${}^{14}_6\text{C} \rightarrow {}^{14}_7\text{C} + {}^0_{-1}\beta$

Nuclear Chemistry (contd.)

Example: The ${}_{-1}^0\beta$ activity of 1g of carbon from a recently cut tree is 0.26 s^{-1} . If the activity of 1 g of carbon in a similar wood sample is 0.16 after some time under the same conditions, determine the storage time.

$$\lambda = 0.693/5730\text{ years} = 1.209 \times 10^{-4}/\text{year}$$

$$N_t = N_0 e^{-\lambda t}$$

$$0.16 = 0.26 e^{-(0.0001209)t}$$

$$\frac{0.16}{0.26} = e^{-(0.0001209)t}$$

$$t = \frac{\ln 0.6154}{-0.0001209} = 4015.7\text{ years}$$

Questions

1. Calculate the moles of ${}_{86}^{222}\text{Rn}$ left after 15.2 days if the initial amount is 90mmol. The half-life of ${}_{86}^{222}\text{Rn}$ is 3.8 days. (Ans.: 5.6 mmol)
2. A bone of an animal was found to emit beta particles from carbon-14. If the activity of the bone was 0.19 s^{-1} and that of a related recent bone of the same animal showed activity of 0.25 s^{-1} for the same mass sample of the bone under the same condition. Determine the age of the bone. Carbon-14 half-life is 5730 years. (Ans.: 2268 years).
3. A rock contains 0.257 mg of ${}_{82}^{206}\text{Pb}$ for every mg of ${}_{92}^{238}\text{U}$. The half-life of the decay of ${}_{92}^{238}\text{U}$ to ${}_{82}^{206}\text{Pb}$ is 4.5×10^9 years. How old is the rock? (Ans.: 1.7×10^9 years)

Nuclear Chemistry (contd.)

Nuclear binding energy and binding energy per nucleon

The binding energy is the energy released during the formation of a nucleus from the appropriate number of neutrons, protons and electrons. Nuclear binding energy is determined by calculating the difference in mass of the nucleus and the sum of the masses of all its subatomic particles. This is called mass defect. The mass defect and energy are related by Einstein theory of mass and energy relationship.

$\Delta E = \Delta m C^2$, ΔE = binding energy, Δm = mass defect, C = speed of light (3×10^8 m/s)

When mass is in atomic mass unit (amu), $1 \text{ amu} = 931 \text{ MeV}$ and $1 \text{ MeV} = 10^6 (1.6 \times 10^{-19}) \text{ V}$.

Binding energy per nucleon is the energy required to pull out a nucleon from the nucleus. it is obtained by dividing the total binding energy by the number of nucleons in the nucleus.

Example: calculate the binding energy of helium, given that it is formed by: $2 {}^1_1\text{H} + 2 {}^1_0\text{n} \rightarrow {}^4_2\text{He}$

$${}^1_1\text{H} = 1.0078 \text{ amu} \quad {}^1_0\text{n} = 1.00087 \text{ amu} \quad {}^4_2\text{He} = 4.0026$$

$$\Delta m = 2(1.0078) + 2(1.00087) - 4.0026 = 0.0304$$

$$\Delta E = 931 \times 0.0304 = 28.3024 \text{ MeV}$$

$$\text{Binding energy per nucleon} = 28.3024/4 = 7.0756 \text{ MeV}$$

Question

1. calculate the energy released during the nuclear reaction : ${}^3_2\text{He} + {}^1_0\text{n} \rightarrow {}^4_2\text{He} + \Delta E$

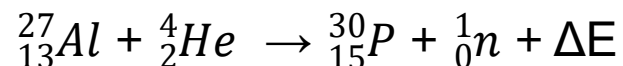
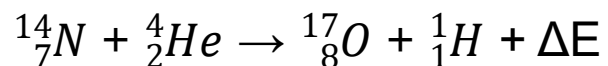
$${}^3_2\text{He} = 3.0160 \text{ amu} \quad {}^1_0\text{n} = 1.00087 \text{ amu} \quad {}^4_2\text{He} = 4.0026 \quad (\text{Ans.: } 20.58 \text{ MeV})$$

1. Calculate the total binding energy and binding energy per nucleon for i. ${}^{24}_{12}\text{Mg}$ with mass = 23.9850 amu and ii. ${}^{60}_{27}\text{Co}$ with mass = 59.9338 amu. Given that ${}^1_1\text{H} = 1.0078 \text{ amu}$ and ${}^1_0\text{n} = 1.00087 \text{ amu}$

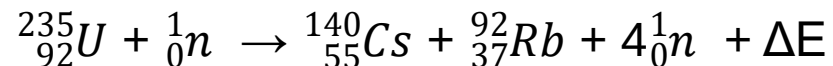
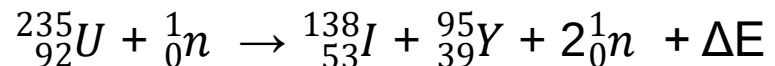
Nuclear Chemistry (contd.)

Nuclear reactions

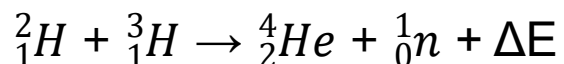
A nuclear reaction can be a spontaneous disintegration, which is associated with unstable nucleus. There can also be induced nuclear reaction, when a stable nucleus is bombarded with a subatomic particle such as beta particle, neutron or alpha particles. This is called artificial radioactivity.



- 1. Nuclear fission** is another induced radioactivity, which involves the breaking down a heavy nucleus into two or more smaller nuclei, when the heavy nucleus is bombarded with a subatomic particle such as neutron. Huge energy is released during this process and it is the reaction behind nuclear power plants. In nuclear power plant, the energy is converted to electricity.



- 2. Nuclear fusion** involves bombardment of two light-weight nuclei to give a relatively heavier nucleus with release of huge energy. This is the process by which solar radiation is generated.



Nuclear Chemistry (contd.)

Detection and measurement of radiation

Devices used in the detection and measurement of radiation depend on the ionization effects of radiation particles and their photographic effect on plate. Examples of the detectors are:

Gas filled detectors: These devices depend on the ionization effect caused by radiation on gas. Examples include; ionization chamber, proportional counter and Geiger-Muller counter.

Scintillation counters: These counters depend on the absorption of radiation by organic or inorganic crystals called phosphor. On absorption of radiation, the phosphor will emit light energy called photon. Examples of organic phosphors are; anthracene and stilbene, while inorganic phosphors include; sodium iodide and zinc sulphide.

Applications of some radioactive isotopes.

Nucleus	Nuclear activity	Application
$^{14}_6\text{C}$	beta emission	Archeological dating
$^{11}_6\text{C}$	positron emission	Brain scan
$^{32}_{15}\text{P}$	beta emission	Detection of breast tumors
$^{15}_8\text{O}$	positron emission	Lung function diagnosis
$^{131}_{53}\text{I}$	beta emission	Measurement of iodine take up by thyroid

Nuclear Chemistry (contd.)

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End of lecture